

The dominant direction of $\mathbf{W}_O$ is not identifiable: gauge orbit, zero certified radius, and consequences for pruning

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Abstract

The value-output factorisation of an attention head is not unique: for every $\mathbf{R} \in GL(d_h)$, the substitution $(\mathbf{W}_v, \mathbf{b}_v, \mathbf{W}_O) \rightarrow (\mathbf{W}_v \mathbf{R}, \mathbf{b}_v \mathbf{R}, \mathbf{R}^{-1} \mathbf{W}_O)$ leaves the function intact, and makes the dominant direction $\mathbf{v}_1(\mathbf{W}_O^{(h)})$ a tempting static proxy for pruning, interpretation or routing. We prove that this readout is not identifiable: its orbit under the gauge is the full unit sphere of the row space, and no admissible similarity threshold on it admits a positive functional certified radius. With a non-degenerate spectrum, only the conformal orthogonal class leaves it invariant for every $\mathbf{W}_O$, and every invariant readout of $\mathbf{W}_O$ alone factors through its row space. The consequence is measured: the pair that weight-based pruning declares most redundant changes under generic reparametrisation in more than 90 % of cases on a ViT and —by the same closed form, without training— on a language transformer. In the gauge training leaves, the factorisation approaches balance without the decisions coinciding. An angular probe that separates the directions up to the simplex threshold produces no detectable functional cost; the hard imposition does. The response signature $\mathbf{v}_1(\mathbf{C}_h^P)$, probe-conditioned and gauge-invariant, measures functional diversity per head. Code, data and demo with DOIs.

Keywords: gauge non-identifiability, OV circuit, head pruning, certified radius, multi-head attention.

1 Introduction

An attention head (Vaswani et al., 2017) writes into the residual stream through the product $\mathbf{W}_v^{(h)} \mathbf{W}_O^{(h)}$, yet it is implemented as two matrices. The discrepancy between the object that acts —the *OV circuit*— and the factors that are stored opens a reparametrisation freedom: for every $\mathbf{R} \in GL(d_h)$, the pair $(\mathbf{W}_v \mathbf{R}, \mathbf{R}^{-1} \mathbf{W}_O)$ implements the same function. That freedom has been known qualitatively since the foundational analysis of attention circuits (Elhage et al., 2021), and the group that generates it is by now characterised: van Nierop (2024) and Hashimoto et al. (2024) formalise it, Wang and Wang (2026) prove its maximality — $((GL(d_k))^h \times (GL(d_v))^h) \rtimes S_h$ on the generic full-rank stratum— and Tran et al. (2026) extend it to positional encodings. What was not written down is the orbit that group induces on a specific *readout* of the weights. The factorisation makes $\mathbf{v}_1(\mathbf{W}_O^{(h)})$ a tempting static proxy —the cheapest imaginable, with no forwards and no gradients, available in the checkpoint— for comparing heads: deciding which are redundant and pruning them, interpreting them by their dominant direction, or routing information between them. The question is whether that readout represents a property of the function or a choice of parametrisation.

This work closes that route with an exact bound. Proposition 1 proves that the orbit of $\mathbf{v}_1(\mathbf{W}_O)$ under the value-output gauge is the full unit sphere of its row space: for any desired direction there

is a reparametrisation that produces it without touching the function. The geometric consequence is that the angle between the dominant directions of two heads sweeps, with the function frozen, the whole interval their row spaces allow. The practical consequence is that no similarity threshold on that direction, within the range the row spaces allow, admits a positive certified radius. The proof fits in a paragraph and the statement in one line: *reading $\mathbf{v}_1(\mathbf{W}_O)$ is reading a gauge choice, not a property of the function.*

The result is fine-grained in two directions, stated from the start. First, the boundary: the subclass $\{c\mathbf{Q}\}$ of scalar multiples of orthogonal matrices —of measure zero in $GL(d_h)$ — leaves $\mathbf{v}_1$ invariant (with a non-degenerate spectrum; with a degenerate head of the spectrum, what is invariant is the dominant subspace), so the freedom to rotate the residual basis and the orthogonal rotations that model compression exploits fall on the harmless side. The impossibility requires the full gauge, and the separation between the two classes is part of the statement, not an afterthought. Second, the object that survives: the row space is gauge-invariant —the direction moves within the subspace, not the subspace—, and the OV circuit stays fixed by construction. The negative result targets every readout of $\mathbf{W}_O$ that does not factor through its row space (corollary 1), not every static reading.

An impossibility about the wrong object would be harmless if the decisions built on it were stable in practice. It was measured: under generic reparametrisation of the sector, the pair of heads that the weight criterion declares most redundant changes in more than 90 % of cases, on a ViT-B/16 finetuned on ImageNet-100 and —applying the same closed form, without training anything— on a 24-layer pretrained language transformer. The impossibility is not a pathology of adversarial gauges: the strength sweep shows the drift growing with the distance to the scalar family while the output error stays at the floor of double precision, twelve orders of magnitude below.

Two natural questions bound the scope of the result, and both are answered empirically. Does training select a benign gauge? In the gauge the optimiser leaves —with none injected— the factorisation moves towards balance (lemma 1): the weight reading aligns with the identifiable object to within 18–35°. The two pruning decisions, however, do not come to coincide, and the most redundant pair differs in 37–44 % of layers. Non-identifiability is not a remote possibility that training sidesteps: it is a margin that training narrows without closing. And how much angular separation does the function tolerate? A probe that separates the dominant directions up to the simplex threshold produces no detectable functional cost at the measured strength. Only the hard imposition —beyond the threshold— pays in function. The coupling between that geometry and the function depends on the mechanism, and the three intervention regimes (free, soft, hard) delimit it.

As a positive instrument, the gauge-invariant response signature $\mathbf{v}_1(\mathbf{C}_h^P)$ —the dominant direction of the second moment of the head’s contribution over a probe set, without per-patch centring— measures functional diversity without the ambiguity, admits an explicit radius against the estimator perturbation (lemma 2), worst-case, and turns out to be head-specific. It is the concrete substitution that practice can adopt where it reads weights today.

Contributions. (i) The exact characterisation of the orbit of $\mathbf{v}_1(\mathbf{W}_O)$ under the value-output gauge —the full sphere of the row space— with its zero certified radius and its orthogonal boundary (section 3); (ii) the numerical certification of the orbit pair by pair at machine precision, 3960/3960 pairs (section 7.1); (iii) the decisional consequence measured in two modalities with the same closed form (section 7.2); (iv) the measurement of training’s natural gauge and of its distance to balance (section 7.3); (v) the contrast of the three intervention regimes and the invariant signature as an instrument (sections 4 and 5.2). Code, data, weights, and an interactive demo are archived with DOIs (data availability section).

2 Related work

The OV circuit as the unit of analysis. That the value-output sector is meaningful only as a product is not this work’s observation. Elhage et al. (2021) organise the analysis of attention into two circuits —QK and OV— and describe the individual matrices as intermediate results: keys, queries and values “can be thought of as intermediate results in the computation of the low-rank matrices $\mathbf{W}_Q^\top \mathbf{W}_K$ and $\mathbf{W}_O \mathbf{W}_V$ ”, so that “it can be useful to describe transformers without reference to them”. The same work states a reparametrisation freedom, that of the residual stream, which has no privileged basis: “we could rotate it by rotating all the matrices interacting with it, without changing model behavior”. This work starts there and adds four pieces absent from that observation. First, the orbit in closed form: not that $\mathbf{v}_1(\mathbf{W}_O)$ is “not unique”, but that its orbit under the sector’s gauge is the full unit sphere of its row space (proposition 1); and rotating the residual does not reach it, because the orthogonal subgroup leaves $\mathbf{v}_1(\mathbf{W}_O)$ fixed and it is the non-orthogonal part —foreign both to that warning and to the rotations of compression— that moves it within the subspace. Second, the impossibility that follows: certified zero radius, no admissible similarity threshold on that direction admits a guarantee. Third, the decisional consequence, measured: the pruning decision changes under gauge in 92.7% of pairs, in vision and in language (table 5). Fourth, an empirical contrast independent of the closed form: the inertness probe moves the geometry by tens of degrees without moving the function (section 7.4). The qualitative observation says which object to look at. This work quantifies what is lost by looking at the other.

The gauge group of the Transformer. The group of reparametrisations that leave the function intact is characterised. Van Nierop (2024) states it by analogy with gauge invariance in particle physics. Hashimoto et al. (2024) unify it with the redundancies of feedforward networks and neural ODEs. Wang and Wang (2026) prove that, for canonical multi-head attention and on the generic stratum of full column rank, the maximal group is exactly $((GL(d_k))^h \times (GL(d_v))^h) \rtimes S_h$, with the query-key and value-output factors factorising independently and head permutation as the only discrete symmetry. Tran et al. (2026) measure how much positional encodings reduce that group. This work takes the factor $(GL(d_v))^h$ of that group —independent per head, on the same full-rank stratum that is a hypothesis here— and does not characterise it: it characterises the orbit it induces on $\mathbf{v}_1(\mathbf{W}_O)$, its uniform stabiliser, the null functional radius and the decision it breaks. Of the group it uses only that this factor is a symmetry —an elementary fact—, not the maximality or the controllability of attention that Wang and Wang require to prove it. The factor S_h lies outside the group studied: under a head permutation the collection of response signatures is equivariant —its labels are permuted—, not a head-by-head invariant family (section 4).

The geometric proxy of the output projection. Reading the role or the redundancy of a head from the dominant direction of its output projection is the cheapest, most static proxy imaginable: it needs no forwards or gradients, it operates directly on the checkpoint. This work closes that route ahead of time more than it corrects a practice already widespread in the reviewed literature. State-of-the-art pruning methods in this setting do not use it: HEART-ViT (Uddin et al., 2025) prunes heads and tokens by Hessian sensitivity on ViT-B/16 on ImageNet-100, Yang et al. (2023) prune with global Hessian saliency, and the head-redundancy analysis with the longest track record reads gradients or activations, not weight geometry (Michel et al., 2019; Voita et al., 2019). The most recent mechanistic interpretability, when it does read weight geometry per head, converges instead towards the gauge-invariant object rather than the isolated dominant direction: Millidge and Black (2022) decompose by SVD the composite OV circuit and the MLP weights, and interpret dozens of singular directions, not just the first; Dar et al. (2023) project into embedding space the products $\mathbf{W}_Q \mathbf{W}_K^\top$ and $\mathbf{W}_V \mathbf{W}_O$, defined explicitly as such; Ahmad et al. (2025) decompose by SVD the composite OV circuit $\mathbf{W}_v \mathbf{W}_O$ (and the QK circuit), not $\mathbf{W}_O$ on its own; Franco et al. (2026) likewise decompose the circuit $\mathbf{W}_q^\top \mathbf{W}_k$; and Yamagiwa

et al. (2026) compare *full-rank* subspaces of the output projection —precisely the object that is here certified invariant under the gauge of the value-output sector (proposition 1)—, not its isolated dominant direction. The correction this work contributes is, accordingly, a corollary of the measured geometry-function dissociation (section 7.4), not the thesis: it protects the cheapest shortcut —reading $\mathbf{v}_1(\mathbf{W}_O)$ alone— that the more careful practice is already moving away from, without claiming to have corrected anyone in particular. Table 1 places those readings against the signature by their properties, with no performance claim whatsoever.

Table 1: Readings of the value-output sector, by properties. Stable by construction if the reading factors through the row space or through the circuit (corollary 1); *measured* only where indicated ($\mathbf{v}_1(\mathbf{W}_O)$ and the signature, table 5). A dash denotes a property not measured here, not an absent one. The harmless class is the conformal orthogonal $\{c\mathbf{Q}\}$, under which $\mathbf{v}_1(\mathbf{W}_O)$ does not move at all.

Reading	Gauge inv.	Data	Per head	Cert. radius	Stable under gauge
$\mathbf{v}_1(\mathbf{W}_O)$	no	no	yes	null (proposition 1)	no (92.7 %)
Dominant subspace of $\mathbf{W}_O$	harmless class	no	yes	—	no
Row space of $\mathbf{W}_O$	yes	no	yes	—	yes
$\mathbf{v}_1(\mathbf{W}_v \mathbf{W}_O)$, static invariant	yes	no	yes	—	yes
Full OV-circuit SVD (Millidge and Black, 2022; Dar et al., 2023; Ahmad et al., 2025)	yes	no	yes	—	yes
Projection kernel between subspaces (Yamagiwa et al., 2026)	yes	no	pairwise	—	yes
Signature $\mathbf{v}_1(\mathbf{C}_h^P)$	yes	yes	yes	positive (lemma 2)	yes (table 5)

Redundancy and roles of multi-head attention. The analysis of multi-head attention has shown that many heads are dispensable (Michel et al., 2019) and that they fulfil specialised, identifiable functions (Voita et al., 2019). A head can be important for the task without the geometry of its output projection tracing that role. Whether that specialisation is an emergent property of training or a static property recoverable from the weights is the question that this work’s results address (sections 7 and 8).

The gauge orbit as a resource, not only a warning. The gauge freedom that is here certified as a source of non-identifiability (proposition 1) is the same one that rotational quantisation actively exploits as a design resource. QuaRot (Ashkboos et al., 2024b) and SpinQuant (Liu et al., 2025) insert orthogonal rotations that leave the model’s output intact —their term, taken from SliceGPT (Ashkboos et al., 2024a), is “computational invariance”— to remove outliers before quantising. SpinQuant’s rotation R_2 is applied exactly to the value-output pair ($\mathbf{W}_v^{(h)} \rightarrow \mathbf{W}_v^{(h)} R_2$, $\mathbf{W}_O^{(h)} \rightarrow R_2^{-1} \mathbf{W}_O^{(h)}$ for every head h of the layer, with R_2 orthogonal), the same gauge of proposition 1 restricted to the orthogonal subgroup $O(d_h) \subset GL(d_h)$ and, unlike the orbit certified here, shared across the heads of the layer rather than independent per head. That restriction is moreover harmless for the reading examined here: under an orthogonal gauge $\mathbf{v}_1(\mathbf{W}_O)$ does not even move (section 3), so rotated compression coexists with the proxy without breaking it, and it is the full gauge —with its non-orthogonal part— that breaks it. That this restriction of the orbit has measurable operational consequences in the compression literature is independent evidence that the choice of gauge is a design space as well as noise to warn against. Quantising the OV circuit along that orbit, and measuring how its error moves with the chosen point, falls outside this work but is the most direct design continuation. Together with component attribution on $Q \cdot K$, it sets the course for future work (section 8). The symmetric question —whether training with weight decay selects a nearly canonical point of that orbit on its own, and whether it suffices— is measured in section 7.3.

Head diversity by regularisation: the direct baseline. Head diversity has been promoted with penalties of several kinds. The closest antecedent intervenes on a signal tied to behaviour: an orthogonality regulariser on the input gradients of each head increases the diversity of the learned features and improves out-of-distribution generalisation (Nicolicioiu et al., 2023). That approach alters the function—and gains OOD from it—, against which this work claims a simpler intervention: angularly separating the *patch-response signature*, with no input gradients or second order, using the same simplex machinery as the discarded weight regulariser. The contrast with intervening on $\mathbf{W}_O$ —which does not alter the function—is consistent with the organisation that distinguishes one head from another residing in the computed response, not in the dominant geometry of the weights.

Depth-wise structure in attention. Functional stratification by depth in Transformers is documented. Pan et al. (2024) decompose the $Q \cdot K$ interaction by SVD of $\mathbf{W}_q^\top \mathbf{W}_k$ and document a transition from perceptual grouping (early layers) to contextualisation (deep ones). Yang et al. (2023), when pruning, observe that the output projection and $Q \cdot K$ have distinct importances and a non-monotonic trend in attention diversity by layer. This depth-wise structure is not a contribution of this work, but context: it frames the seam at layer 5, where the organisation of attention leaves residual writing and moves to the query-key comparison, and the inversion towards $Q \cdot K$ that we document (section 7) is a neighbour of those observations.

Neural collapse, fixed classifiers and angular margins. *Neural collapse* (Papayan et al., 2020; Zhu et al., 2021) describes the convergence of class means and classifier to a tight equiangular frame and has motivated *fixed* classifiers of predetermined geometry (Pernici et al., 2022); SphereFace (Liu et al., 2017), CosFace (Wang et al., 2018) and ArcFace (Deng et al., 2019) impose angular margins on prototypes estimated during training. Here the simplex is a tool of the experiment, not a contribution: it fixes a guaranteed separation on each head’s response signature, not on its weights.

3 The value-output gauge and the zero certified radius

The practice this work examines reads one direction of the weights: the first singular vector $\mathbf{v}_1(\mathbf{W}_O^{(h)})$ of each head’s output projection, taken as its representative direction for redundancy pruning, dominant-direction interpretation, or routing. This section fixes what of that sector is identifiable and what is not. The instrument that rests on the identifiable object is derived in section 4.

A head’s value-output sector factorises its contribution to the residual stream into two matrices, $\mathbf{W}_v^{(h)} \in \mathbb{R}^{d \times d_h}$ and $\mathbf{W}_O^{(h)} \in \mathbb{R}^{d_h \times d}$, but that factorisation is not unique. For any invertible $\mathbf{R} \in GL(d_h)$, the substitution

$$\mathbf{W}_v^{(h)} \rightarrow \mathbf{W}_v^{(h)} \mathbf{R}, \quad \mathbf{b}_v^{(h)} \rightarrow \mathbf{b}_v^{(h)} \mathbf{R}, \quad \mathbf{W}_O^{(h)} \rightarrow \mathbf{R}^{-1} \mathbf{W}_O^{(h)}, \quad (1)$$

leaves the function intact, because the product $\mathbf{W}_v^{(h)} \mathbf{W}_O^{(h)}$ —the *OV circuit*—and the bias term $\mathbf{b}_v^{(h)} \mathbf{W}_O^{(h)}$ do not change. Without a value bias the invariant is the circuit alone, and with it it is the augmented matrix $[\mathbf{W}_v^{(h)}; \mathbf{b}_v^{(h)}] \mathbf{W}_O^{(h)}$. Everything that follows holds equally in both cases, because it uses only that $\mathbf{W}_v \mathbf{W}_O$ is invariant. Within the value-output factorisation, functional properties invariant under the gauge must descend to the OV circuit. The dominant direction $\mathbf{v}_1(\mathbf{W}_O^{(h)})$ does not descend. Under eq. (1) it shifts, and the part of the group that moves it can be identified exactly. If $\mathbf{R}$ is orthogonal—or a scalar multiple of an orthogonal one, $\mathbf{R}^{-1} = c\mathbf{Q}$ with $\mathbf{Q} \in O(d_h)$ —, then $\mathbf{R}^{-1} \mathbf{W}_O = (\mathbf{Q}\mathbf{U})(c\mathbf{\Sigma})\mathbf{V}^\top$ is a singular value decomposition: $\mathbf{Q}\mathbf{U}$ is orthonormal and the scalar is absorbed into the singular values—with $c > 0$ without loss, since the sign is absorbed into $\mathbf{Q}$ —, so the right singular vectors are the same and the direction does

not move. The invariance is of the vector while $\sigma_1 > \sigma_2$. With a degenerate head of the spectrum, what is invariant is the dominant subspace. The converse holds uniformly in $\mathbf{W}_O$: if $\mathbf{S} = \mathbf{R}^{-1}$ leaves $\mathbf{v}_1(\mathbf{S}\mathbf{W}) = \pm \mathbf{v}_1(\mathbf{W})$ for every full-rank $\mathbf{W}$ with $\sigma_1 > \sigma_2$, then $\mathbf{S} = c\mathbf{Q}$. With $\mathbf{W} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ one has $\mathbf{v}_1(\mathbf{S}\mathbf{W}) = \mathbf{V}\mathbf{v}_1(\mathbf{S}\mathbf{U}\mathbf{\Sigma})$, and the condition requires $\mathbf{e}_1$ to be the dominant eigenvector of $\mathbf{\Sigma}\mathbf{U}^\top\mathbf{S}^\top\mathbf{S}\mathbf{U}\mathbf{\Sigma}$ for every $\mathbf{U} \in O(d_h)$ and every admissible $\mathbf{\Sigma}$. With $\mathbf{\Sigma} = \mathbf{I} + \varepsilon\mathbf{e}_1\mathbf{e}_1^\top$ and $\varepsilon \rightarrow 0$, that eigenvector approaches the dominant eigenspace $\mathbf{U}^\top\mathcal{E}$ of $\mathbf{U}^\top\mathbf{S}^\top\mathbf{S}\mathbf{U}$, with $\mathcal{E}$ that of $\mathbf{S}^\top\mathbf{S}$; if $\mathbf{S}^\top\mathbf{S} \neq c^2\mathbf{I}$, $\mathcal{E}$ is a proper subspace, a $\mathbf{U}$ with $\mathbf{U}\mathbf{e}_1 \notin \mathcal{E}$ leaves $\mathbf{e}_1$ at positive distance from $\mathbf{U}^\top\mathcal{E}$, and the condition fails for small ε . Hence $\mathbf{S}^\top\mathbf{S} = c^2\mathbf{I}$ and $\mathbf{S} = c\mathbf{Q}$. Uniqueness is uniform, not pointwise: on a specific $\mathbf{W}$, the stretch $\mathbf{S} = \mathbf{I} + (a-1)\mathbf{u}_1\mathbf{u}_1^\top$ along its first left singular vector, with $a > 1$, also leaves $\mathbf{v}_1$ fixed without being conformal orthogonal. It is the non-orthogonal part, present in every generic gauge, that moves the direction within the row space. The freedom to rotate the residual basis (Elhage et al., 2021) and the orthogonal rotations of compression (Ashkboos et al., 2024a,b; Liu et al., 2025) therefore fall on the harmless side: the non-identifiability certified here requires the full gauge. With it, two functionally identical models can exhibit any angular separation between their $\mathbf{v}_1(\mathbf{W}_O)$. The claim admits a closed form and an exact bound:

The proposition’s title uses a term that is fixed before stating it. Given a readout f of the sector with threshold decision d_τ , and a perturbation class $(\mathcal{P}, \text{dist})$ with $\text{dist}(\text{id}) = 0$, the local *certified radius* at a point θ of the sector is the supremum of the r such that every $P \in \mathcal{P}$ with $\text{dist}(P) \leq r$ leaves $d_\tau(f(P \cdot \theta)) = d_\tau(f(\theta))$. For the sector’s gauge, the natural distance is the *singular-spectrum spread*,

$$d(\mathbf{R}) = \min_{c>0, \mathbf{Q} \in O(d_h)} \frac{\|\mathbf{R} - c\mathbf{Q}\|_F}{\|\mathbf{R}\|_F} = \frac{(\sum_i (\sigma_i - \bar{\sigma})^2)^{1/2}}{(\sum_i \sigma_i^2)^{1/2}}, \quad (2)$$

zero exactly on the *harmless class* $\{c\mathbf{Q}\}$ that leaves $\mathbf{v}_1$ fixed (the optimal $\mathbf{Q}$ and c : those of the polar decomposition and the mean $\bar{\sigma}$ of the singular values of $\mathbf{R}$). A certificate about the *function* further requires *soundness over the orbit*: $d_\tau \circ f$ must be constant on the gauge orbit of θ , whose points all implement the same function. The *functional* certified radius is then dichotomous, not graded: it equals the local radius if the readout descends to the quotient by the orbit, and is zero if it does not. The dichotomy is not vacuous because it separates objects: the row space, the OV circuit, and the response signature descend—they fall on the positive side, and for the signature section 4 further gives a positive local radius against the estimator perturbation—. The dominant direction does not descend, and that is all the proposition proves.

Proposition 1 (Null certified radius of the $\mathbf{v}_1$ readout). *Let $\mathbf{W}_O^{(h)}$ be a full-rank matrix in $\mathbb{R}^{d_h \times d}$, and let $\mathcal{S}_h$ be its row space. For every unit direction $\mathbf{w} \in \mathcal{S}_h$ there exists $\mathbf{R} \in GL(d_h)$ with $\mathbf{v}_1(\mathbf{R}^{-1}\mathbf{W}_O^{(h)}) = \pm \mathbf{w}$, and the substitution (1) leaves the OV circuit intact. Consequently, the angle between the dominant directions of two heads sweeps, under gauge, every angle allowed by the row spaces—from the first principal angle $\theta_1(\mathcal{S}_h, \mathcal{S}_{h'})$ up to 90° —with the function fixed, and no similarity threshold on $\mathbf{v}_1(\mathbf{W}_O)$ within the admissible interval $(\theta_1(\mathcal{S}_h, \mathcal{S}_{h'}), 90^\circ)$ admits a positive certified radius.*

Proof. With $\mathbf{W}_O = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and $\mathbf{M} = \mathbf{R}^{-1}\mathbf{U}\mathbf{\Sigma}$ we have $\mathbf{R}^{-1}\mathbf{W}_O = \mathbf{M}\mathbf{V}^\top$: when the top spectrum of $\mathbf{M}$ is simple ($\sigma_1(\mathbf{M}) > \sigma_2(\mathbf{M})$), the first right singular vector of $\mathbf{M}$ is well defined up to sign and $\mathbf{v}_1(\mathbf{R}^{-1}\mathbf{W}_O) = \mathbf{V}\mathbf{v}_1(\mathbf{M})$. Given a unit $\mathbf{w} \in \mathcal{S}_h$, let $\mathbf{w}' = \mathbf{V}^\top\mathbf{w} \in \mathbb{R}^{d_h}$ be its expression in the row-space basis and $\mathbf{M} = \mathbf{I} + (a-1)\mathbf{w}'\mathbf{w}'^\top$ with $a > 1$: symmetric, with singular values $\{a, 1, \dots, 1\}$, simple top by construction, and $\mathbf{v}_1(\mathbf{M}) = \mathbf{w}'$. The transformation $\mathbf{R} = \mathbf{U}\mathbf{\Sigma}\mathbf{M}^{-1} \in GL(d_h)$ —invertible because $\mathbf{\Sigma}$ is, by the full rank—realises that $\mathbf{M}$, and then $\mathbf{v}_1(\mathbf{R}^{-1}\mathbf{W}_O) = \mathbf{V}\mathbf{w}' = \pm \mathbf{w}$. The orbit of $\mathbf{v}_1$ is the full unit sphere of $\mathcal{S}_h$. With independent gauges per head, the pair of dominant directions sweeps the product of the two spheres, which is connected, and the angle between them—continuous—takes every value between its minimum, $\theta_1(\mathcal{S}_h, \mathcal{S}_{h'})$ by definition of the first principal angle, and 90° , attained by two principal vectors of distinct index ($d_h \geq 2$ suffices). For every threshold in the open interval, $d_\tau \circ f$ is not constant on the orbit: the readout does not descend to the quotient and its functional radius is zero. $\square$

proposition 1 does not restate the qualitative observation that the OV circuit is the unit of analysis (Elhage et al., 2021): it turns it into a bound. It names the orbit —the full sphere of $\mathcal{S}_h$ —, separates it from the already known orthogonal freedom —under which $\mathbf{v}_1(\mathbf{W}_O)$ is invariant— and from there extracts the certified zero radius: for every threshold within the admissible interval, the decision $d_\tau \circ f$ is not constant on the orbit —the angle sweeps it whole—, so the readout does not descend to the quotient and its functional radius is zero, with no local construction at all. It is the step from warning to impossibility, and each half of the formalism earns its keep in a different place: on the negative side, the radius reformulates the proposition. On the positive one —the signature— it delivers an explicit number.

The nuance that makes this non-trivial: the row space *is* gauge invariant. What moves is the direction within the subspace, not the subspace. The negative result is therefore fine-grained —the subspace does carry information (appendix B.1); the dominant direction within it does not—. Reading $\mathbf{v}_1$ is reading a choice of gauge, not a property of the function. Section 7.1 measures this on the real weights: the separation between the $\mathbf{v}_1(\mathbf{W}_O)$ shifts with $\mathbf{R}$ while the output and the OV circuit remain fixed, and the attainable orbit is certified pair by pair (appendix A).

The proposition admits a more general reading, which names what information of $\mathbf{W}_O$ alone is identifiable:

Corollary 1 (Quotient of $\mathbf{W}_O$ by the gauge). *Let $\mathbf{W}_O \in \mathbb{R}^{d_h \times d}$ have full row rank and let $\mathcal{S}_h$ be its row space. The orbit $\{\mathbf{R}^{-1}\mathbf{W}_O : \mathbf{R} \in GL(d_h)\}$ is exactly the set of rank- d_h matrices with row space $\mathcal{S}_h$, so that the quotient is the Grassmannian, $\{\mathbf{W} \in \mathbb{R}^{d_h \times d} : \text{rank } \mathbf{W} = d_h\} / GL(d_h) \simeq \text{Gr}(d_h, d)$, and every gauge-invariant readout of $\mathbf{W}_O$ alone factors through $\mathcal{S}_h$. For the pair $(\mathbf{W}_v, \mathbf{W}_O)$ of full ranks, the complete invariant is the circuit $\mathbf{C} = \mathbf{W}_v \mathbf{W}_O$.*

Proof. If $\mathbf{W}'$ has rank d_h and row space $\mathcal{S}_h$, its rows are invertible combinations of those of $\mathbf{W}_O$: $\mathbf{W}' = \mathbf{T}\mathbf{W}_O$ with $\mathbf{T} = \mathbf{W}'\mathbf{W}_O^+$ unique and invertible, hence $\mathbf{W}' = \mathbf{R}^{-1}\mathbf{W}_O$ with $\mathbf{R} = \mathbf{T}^{-1}$. The converse is immediate because $\mathbf{R}^{-1}$ is invertible. An invariant function is constant on each orbit, that is, on each fibre of $\mathbf{W}_O \mapsto \mathcal{S}_h$, and therefore factors through it. For the pair: if $\mathbf{W}_v \mathbf{W}_O = \mathbf{W}_v' \mathbf{W}_O'$ with full ranks, the column spaces of $\mathbf{W}_v$ and $\mathbf{W}_v'$ coincide with that of $\mathbf{C}$, hence $\mathbf{W}_v' = \mathbf{W}_v \mathbf{R}$ with $\mathbf{R} = \mathbf{W}_v^+ \mathbf{W}_v'$ invertible, and $\mathbf{W}_v(\mathbf{R}\mathbf{W}_O' - \mathbf{W}_O) = \mathbf{0}$ with $\mathbf{W}_v$ of full column rank gives $\mathbf{W}_O' = \mathbf{R}^{-1}\mathbf{W}_O$. $\square$

The corollary places the readouts of table 1: the row space and the projection kernel between subspaces (Yamagiwa et al., 2026) factor through $\text{Gr}(d_h, d)$ and are invariant by construction. The dominant direction and the dominant subspace of $\mathbf{W}_O$ do not factor and are not. Proposition 1 is the extreme case: the readout not only fails to factor, its orbit exhausts the sphere of the fibre.

It remains to name the point of the orbit that training itself might prefer, because section 7.3 measures it on the real weights. The value bias does not enter the penalty and does not alter the lemma:

Lemma 1 (Balanced point). *Let $\mathbf{C} = \mathbf{W}_v^{(h)} \mathbf{W}_O^{(h)}$ have full rank d_h . Over $GL(d_h)$, the minimum of $\|\mathbf{W}_v \mathbf{R}\|_F^2 + \|\mathbf{R}^{-1}\mathbf{W}_O\|_F^2$ at fixed circuit equals $2\|\mathbf{C}\|_*$ —the nuclear norm— and is attained exactly at the balanced factorisations, $(\mathbf{W}_v \mathbf{R})^\top (\mathbf{W}_v \mathbf{R}) = (\mathbf{R}^{-1}\mathbf{W}_O)(\mathbf{R}^{-1}\mathbf{W}_O)^\top$, unique up to the orthogonal subgroup $O(d_h)$. If moreover $\sigma_1(\mathbf{C}) > \sigma_2(\mathbf{C})$, at every balanced point $\mathbf{v}_1(\mathbf{W}_O) = \pm \mathbf{v}_1(\mathbf{C})$: the weight readout coincides with the invariant. With a degenerate spectrum the coincidence is of the dominant subspace, the same clause that proposition 1 carries for $\mathbf{W}_O$.*

Proof. For every factorisation $\mathbf{C} = \mathbf{A}\mathbf{B}$ with $\mathbf{A} = \mathbf{W}_v \mathbf{R}$, $\mathbf{B} = \mathbf{R}^{-1}\mathbf{W}_O$: $2\|\mathbf{C}\|_* \leq 2\|\mathbf{A}\|_F \|\mathbf{B}\|_F \leq \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2$ —the first inequality is Hölder for the Schatten norms with $p=q=2$ (Bhatia, 1997), the second is AM–GM; together they are the variational form of the nuclear norm (Srebro and Shraibman, 2005)—. With the reduced $\mathbf{C} = \mathbf{U}\Sigma\mathbf{V}^\top$, the pair $\mathbf{A}_0 = \mathbf{U}\Sigma^{1/2}$, $\mathbf{B}_0 = \Sigma^{1/2}\mathbf{V}^\top$ attains the value and is balanced. It is moreover reachable within the orbit: $\mathbf{R} = \mathbf{W}_v^+ \mathbf{A}_0$ satisfies $\mathbf{W}_v \mathbf{R} = \mathbf{A}_0$ —the columns of $\mathbf{A}_0$ live in the column space of $\mathbf{C}$, contained in that of $\mathbf{W}_v$ — and is invertible because $\mathbf{A}_0$ has rank d_h . The identity $\mathbf{W}_v(\mathbf{R}\mathbf{B}_0 - \mathbf{W}_O) = \mathbf{0}$ with $\mathbf{W}_v$ of full column

rank gives $\mathbf{R}^{-1}\mathbf{W}_O = \mathbf{B}_0$. That there are no other minimisers follows from stationarity. The function $\varphi(\mathbf{R}) = \|\mathbf{W}_v\mathbf{R}\|_F^2 + \|\mathbf{R}^{-1}\mathbf{W}_O\|_F^2$ is smooth on the open set $GL(d_h)$ and coercive on it— $\|\mathbf{W}_v\mathbf{R}\|_F \geq \sigma_{\min}(\mathbf{W}_v)\|\mathbf{R}\|_F$ and $\|\mathbf{R}^{-1}\mathbf{W}_O\|_F \geq \sigma_{\min}(\mathbf{W}_O)\|\mathbf{R}^{-1}\|_F$, both $\sigma_{\min}$ positive by the full rank—, so every minimiser is a stationary point. Its gradient is $2\mathbf{W}_v^\top\mathbf{W}_v\mathbf{R} - 2\mathbf{R}^{-\top}\mathbf{B}\mathbf{B}^\top$; setting it to zero and multiplying by $\mathbf{R}^\top$ on the left gives $\mathbf{A}^\top\mathbf{A} = \mathbf{B}\mathbf{B}^\top$: every minimiser is balanced. From the balance and $\mathbf{C} = \mathbf{A}\mathbf{B}$ it follows that $\mathbf{C}^\top\mathbf{C} = \mathbf{B}^\top(\mathbf{A}^\top\mathbf{A})\mathbf{B} = (\mathbf{B}^\top\mathbf{B})^2$, hence $\mathbf{B}^\top\mathbf{B} = (\mathbf{C}^\top\mathbf{C})^{1/2} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^\top$ —the unique semidefinite root—and $\mathbf{B} = \mathbf{Q}^\top\mathbf{\Sigma}^{1/2}\mathbf{V}^\top$ with $\mathbf{Q} \in O(d_h)$; then $\mathbf{A} = \mathbf{C}\mathbf{B}^\top(\mathbf{B}\mathbf{B}^\top)^{-1} = \mathbf{U}\mathbf{\Sigma}^{1/2}\mathbf{Q}$. That whole family attains $2\|\mathbf{C}\|_*$ and there are no minimisers outside it. Within it, the right singular vectors of $\mathbf{B}$ are those of $\mathbf{C}$, and with $\sigma_1 > \sigma_2$ the first is $\mathbf{v}_1(\mathbf{C})$ up to sign. $\square$

Numerical verification, with no forwards: over fifty random instances the SVD construction reconstructs the circuit, satisfies the balance, and no perturbation of $\mathbf{R}$ improves it. On the anchor, the angle α between the weight readout and the invariant, evaluated at the constructed balanced point, is exact numerical zero on all 144 heads. The point the lemma names is the reference against which section 7.3 reads the 18–35° of the gauge training leaves.

The balanced point is not the only canonical form of the orbit. Wang and Wang (2026) fix the value-output gauge by a QR factorisation of $\mathbf{W}_v$ —orthonormal columns, residual $O(d_h)$ per head—and name the Gram balance as an alternative in the query-key sector. On the QR form too $\mathbf{v}_1(\mathbf{W}_O) = \pm\mathbf{v}_1(\mathbf{C})$: with $\mathbf{W}_v = \mathbf{U}\mathbf{Q}$ one has $\mathbf{W}_O = \mathbf{W}_v^\top\mathbf{C} = \mathbf{Q}^\top\mathbf{\Sigma}\mathbf{V}^\top$, and the right singular vectors are those of $\mathbf{C}$. More generally, every gauge with $\mathbf{W}_v = \mathbf{U}\mathbf{D}\mathbf{Q}$ — $\mathbf{D}$ diagonal positive in the left singular basis of $\mathbf{C}$, with σ_1/d_1 maximal—gives $\mathbf{W}_O = \mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{\Sigma}\mathbf{V}^\top$ and aligns the readout with the invariant. The residual $O(d_h)$ of the gauge fixing is the orthogonal part of the harmless class $\{c\mathbf{Q}\}$, under which the readout does not move. What the lemma adds is which representative of that family training favours: the one of minimal Frobenius norm, towards which *weight decay* pushes, and that is the reference of section 7.3. The angle α measured there is zero over the whole aligned family, so the reading does not depend on which of its canonical forms is named.

4 The gauge-invariant response signature

If the dominant direction is not a functionally identifiable representation of the head, a measure of head diversity must be anchored to an identifiable object. The head’s per-patch contribution to the residual stream,

$$\mathbf{C}_h^P = \mathbf{A}_h\mathbf{V}_h\mathbf{W}_O^{(h)} \in \mathbb{R}^{P \times d}, \quad (3)$$

—the product of the attention map $\mathbf{A}_h$, the value projection $\mathbf{V}_h$, and the output projection $\mathbf{W}_O^{(h)}$ —is the response the head *computes* and writes into the residual stream, and it is gauge-invariant: $\mathbf{A}_h\mathbf{V}_h\mathbf{R}\mathbf{R}^{-1}\mathbf{W}_O^{(h)} = \mathbf{C}_h^P$. Its first right singular vector,

$$\mathbf{r}_h = \mathbf{v}_1(\mathbf{C}_h^P) \in \mathbb{S}^{d-1}, \quad (4)$$

—the probe-conditioned *response signature*, $\mathbf{r}_h = \mathbf{r}_h(P)$, the residual-stream direction in which the head deposits the most energy across patches—is well-defined across heads and lives in the shared residual stream $\mathbb{R}^{768}$, the same space as $\mathbf{v}_1(\mathbf{W}_O)$, which makes the weight readout and the function readout comparable.¹ The exact invariance is against the sector’s continuous gauge; under a head permutation the collection $\{\mathbf{r}_h\}$ is equivariant, not invariant. The signature depends on the probe—on its size and on its class composition (lemma 2)—. The gauge changes the parametrisation without changing the function, the probe chooses which functional distribution is summarised, and the two choices are not of the same kind. This signature is the work’s measurement instrument: it

¹The singular vector is defined up to sign. Comparisons between singular signatures therefore use the absolute value of the dot product.

quantifies heads’ functional diversity without the gauge ambiguity that invalidates the $\mathbf{v}_1(\mathbf{W}_O)$ readout. Appendix B.3 contrasts it with the static readout, and appendix B.2 verifies that it is head-specific by matching across probe halves.

Against the gauge, the signature needs no radius: it is exactly invariant. Against the estimator perturbation it does admit one, and with an observable gap:

Lemma 2 (Radius of the signature against the estimator). *Let $\hat{\mathbf{C}}_h$ be a matrix in $\mathbb{R}^{P \times d}$ with $\sigma_1 > \sigma_2$, and let $\mathbf{E} \in \mathbb{R}^{P \times d}$ satisfy $\|\mathbf{E}\|_2 < \sigma_1 - \sigma_2$. Then*

$$\sin \theta(\mathbf{v}_1(\hat{\mathbf{C}}_h), \mathbf{v}_1(\hat{\mathbf{C}}_h + \mathbf{E})) \leq \frac{\sqrt{2} \|\mathbf{E}\|_2}{\sigma_1 - \sigma_2 - \|\mathbf{E}\|_2},$$

and if a layer’s signature decision has margin m —the difference in $|\cos|$ between the winning pair and the runner-up— and g is the minimum gap over the layer’s heads —any head can enter a competing pair, and a distant pair with small gaps would escape a minimum restricted to the winner and the runner-up—, the decision is invariant under every per-head perturbation with $\|\mathbf{E}\|_2 < gm/(8 + m)$. The radius is positive for every gap $g > 0$ and every margin $m > 0$; with a tie ($m = 0$) it is zero.

Proof. By Weyl (Stewart and Sun, 1990), the perturbed separation is at least $\sigma_1 - \sigma_2 - \|\mathbf{E}\|_2$, and the theorem of Wedin (1972) with that separation, in the joint form over the two singular subspaces —whence the factor $\sqrt{2}$ (Chen et al., 2021, Theorem 2.9)—, gives the sine bound. The chord bounds the vector’s displacement, $\|\Delta \mathbf{v}\| = 2 \sin(\theta/2) \leq \sqrt{2} \sin \theta$ for $\theta \leq 90^\circ$, so per head $\|\Delta \mathbf{v}\| \leq 2\|\mathbf{E}\|_2/(g - \|\mathbf{E}\|_2)$; each pairwise $|\cos|$ moves at most the sum of its two heads, because for unit vectors $\langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{u}', \mathbf{w}' \rangle = \langle \mathbf{u} - \mathbf{u}', \mathbf{w} \rangle + \langle \mathbf{u}', \mathbf{w} - \mathbf{w}' \rangle$. Every pair other than the winner has deficit at least m —the runner-up realises the maximum—, and its rise, like the winner’s fall, is bounded by the gaps of its own heads; with g the layer minimum, the joint erosion of any competitor against the winner stays below m while $8\|\mathbf{E}\|_2/(g - \|\mathbf{E}\|_2) < m$, which solves to the radius in the statement. $\square$

The verification’s pre-registration —fixed before reading any number, with its dated derivation amendment— declared the radius non-vacuous only if it exceeds the estimator’s real fluctuation, measured as the half-probe swap. At $P=1000$ it does not: the median of $r_h/\|\mathbf{E}\|$ over the anchor’s 144 heads is 0.001, and in 134 the fluctuation exceeds the gap itself, voiding the certificate by construction. The radius is positive and computable, but worst-case: it does not cover the half-probe swap, the coarsest perturbation in its class. A diagnostic outside the pre-registration completes the reading with the real perturbation, not the adversarial one: recomputing the signature on each half of the probe separately, the most redundant pair coincides across halves in 8 of the anchor’s 12 layers (likewise the top-3 pruning set). The signature’s exact stability is against the gauge. Against the estimator it is a finite-sample matter, and at this P the signature decision retains sampling sensitivity in a third of the layers.

That 8 out of 12 is one point on a curve, measured over nested probes —each size a prefix of the next, in class-rotated order— with $P \in \{250, 500, 1000, 2000, 5000\}$ and the anchor’s five seeds. Pair coincidence across halves rises from 10.4 ± 0.9 to 11.4 ± 0.5 over the 12 layers (mean and standard deviation across seeds) on going from 250 to 5000 images, without reaching 12 out of 12 in any sustained seed —two of five reach it at $P=5000$ and none falls below 11—, and the top-3 set rises from 7.8 to 10.4. The median of $r_h/\|\mathbf{E}\|$, by contrast, does not move with size: it stays between 0.0014 and 0.0017 across the grid, with 89-90 % of heads out of domain at every P . Enlarging the probe improves the empirical stability of the decision and does not bring the worst-case certificate closer to the fluctuation: they denote different things, and the curve separates them. That 8 out of 12 also admits a compositional refinement: the halves of the published probe are blocks of disjoint classes, and redoing the same $P=1000$ with class-balanced halves —the same thousand images— raises coincidence to 10 out of 12. Two of the four discrepant layers were so by class composition, not by sample size.

5 Three intervention regimes on the value-output sector

This work intervenes on the value-output sector in three intervention regimes: the free gauge of section 3, which by construction does not touch the function; an angular probe that trains while separating the dominant directions up to a threshold with a guarantee; and a hard variant that imposes the target configuration at every step. This section fixes the angular target —the simplex of the loose regime— and defines the probe and its variant.

5.1 Spherical codes and separation regimes

Definition 1 (Spherical code). *A spherical code of size K in dimension d is a set $\mathcal{C} = \{\mathbf{u}_1, \dots, \mathbf{u}_K\} \subset \mathbb{S}^{d-1}$ of unit directions. Its minimum angular separation is*

$$\theta_{\min}(\mathcal{C}) = \min_{i \neq j} \arccos(\langle \mathbf{u}_i, \mathbf{u}_j \rangle). \quad (5)$$

The goal is to find $\mathcal{C}^*$ that maximises $\theta_{\min}$. In the loose regime ($K \leq d + 1$), the only one this work visits ($K=12$ or 16 against $d=768$ or 1024), the signed optimum is the regular simplex and is closed-form: its Gram matrix factorises —ones on the diagonal, $-1/(K - 1)$ off it— and embeds in $\mathbb{R}^d$ by rotation. The hard variant’s target code (section 5.2) is that configuration. Its construction is deterministic and reproducible without any descent.

The angular separation attainable by a code of K directions in $\mathbb{S}^{d-1}$ depends on the relation between K and d , and on whether the directions carry a sign. For signed directions, summing the K unit directions and observing that the squared norm is non-negative,

$$0 \leq \left\| \sum_h \mathbf{u}_h \right\|^2 = K + \sum_{h \neq h'} \langle \mathbf{u}_h, \mathbf{u}_{h'} \rangle, \quad (6)$$

the mean pairwise cosine is at least $-1/(K - 1)$, and the maximum cosine cannot fall below its mean: $\theta_{\min} \leq \arccos(-1/(K - 1))$ for *any* configuration, not only the equiangular ones. In the *loose regime* ($K \leq d + 1$) the bound is attained —with equality exactly when the sum vanishes and the configuration is equiangular— by the regular simplex, which is therefore optimal (Rankin, 1955). For directions defined *up to sign* —singular vectors, elements of projective space— the relevant similarity is $|\cos|$ and the unsigned angle lives in $[0^\circ, 90^\circ]$: if $K \leq d$, the maximum is attained by the orthogonal frame (90°), and the simplex read without sign ($\arccos(1/(K - 1))$) falls below it. The role of each value in the regulariser —signed optimum, unsigned threshold— is made precise in section 5.2. The 84.78° is therefore not the optimum of the problem the probe regularises —that is the orthogonal frame— but an operational threshold inherited from the simplex. That the soft probe overshoots it and continues towards the orthogonal frame (section 7.4) is what one expects. The saturated regime ($K > d + 1$) is not visited in this work.

5.2 The regulariser on $\mathbf{W}_O$ as an inertness probe

To test whether intervening on the weight geometry moves the function, this work uses as a *probe* —not as a proposed method— the natural angular regulariser built on that proxy: separating the representative directions $\{\mathbf{v}_1(\mathbf{W}_O^{(h)})\}_{h=1}^H$ towards the simplex threshold. The set is a spherical code of H directions defined up to sign, and the term pushes their absolute cosines below the threshold,

$$\mathcal{R}_{\text{ov}} = \sum_{h \neq h'} \max(0, |\langle \mathbf{v}_1(\mathbf{W}_O^{(h)}), \mathbf{v}_1(\mathbf{W}_O^{(h')}) \rangle| - \cos \theta^*), \quad (7)$$

with $\theta^* = \arccos(1/(H - 1)) \approx 84.78^\circ$ the threshold inherited from the H -vector simplex (section 5.1). The absolute value is necessary: two heads with nearly antiparallel directions are equivalent up to sign, and the signed product would read them as well separated. The penalty

vanishes across the whole interval $|\cos| \leq 1/(H-1)$, so it demands sufficient separation, not a specific configuration. The probe is added to the cross-entropy with weight λ ,

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{R}_{\text{ov}}, \quad (8)$$

and the empirical question is whether, upon separating those directions, the functional similarity between heads moves. Section 7.4 detects no shift of functional similarity above between-seed variability at the fixed intensity ($\lambda = 10^{-3}$).

6 Experimental setup

Data set. We use ImageNet-100, the 100-class subset of ImageNet defined by the CMC list (Tian et al., 2020). A hundred classes provide enough angular richness for between-head diversity to be measurable, and the phenomenon under study depends on the relation between the number of heads and the dimension, not on the number of classes. ImageNet-100 is thus representative without the full scale.

Architectures and environment. The apparatus—the $GL(d_h)$ gauge of the value-output sector, the OV circuit, the $Q \cdot K$ interaction, and the $\mathcal{C}_h^P$ signature—is defined the same way across any global attention, so it is contrasted across three columns that vary the two relevant axes without reinterpreting a single metric (table 2). ViT-B/16 and ViT-L/16 (Dosovitskiy et al., 2021) change the scale and depth of the backbone at constant per-head width (H from 12 to 16, d from 768 to 1024, L from 12 to 24, with $d_h = 64$ and hence $H/d = 1/64$ fixed); DINOv2 (Oquab et al., 2024) —ViT-L/14, same width and depth— changes the training regime—self-supervised versus supervised—at that same scale. All three start from pretrained weights (timm 1.0.22; Wightman, 2019). ViT-B and ViT-L are finetuned on ImageNet-100, and DINOv2 is analysed frozen, with no finetuning at all, so its attention maps are those of pure self-supervised pretraining. The anchor architecture (ViT-B) is trained with **five seeds** ($\{42, \dots, 46\}$) and ViT-L with three. The frozen column is a single realisation (a single pre-trained checkpoint; no between-seed variability is estimated) and its cells carry no band. n is reported per row. The contrast tests transfer between two scales and depths at constant d_h and, in the ViT-L column, between pre-training regimes; it does not vary the heads-to-dimension quotient independently. Between-seed variability is the noise floor against which effects are contrasted. The environment is a single 32 GB NVIDIA RTX 5090 GPU. Windowed attention—Swin and, in general, the non-global kind—is left out: non-identifiability would transfer (the value-output gauge is structural to any value-output factorisation), but inertness and relocation are measured on objects that its hierarchical, local topology redefines—the map stops being global, depth stops being uniform—and they enter as an extension, not as a column of these tables.

Table 2: Contrasted architectures and outcome by support. ViT-B→ViT-L varies scale and depth at constant d_h and H/d ; ViT-L→DINOv2 varies the regime at constant scale. n : seeds; the frozen column is a single realisation, with no between-seed spread. Gauge: output invariance (fp64) with drift of $\mathbf{v}_1(\mathbf{W}_O)$; residual: layers with paired $< \text{null}$; crossover: layer of the sustained crossover $r_{Q \cdot K} > r_{\mathbf{W}_O}$ against the 24-layer thresholds fixed before the data; inertness (H1): only finetuned columns. ViT-L’s crossover passes two of the three thresholds (jump +0.403, margin +0.366) and fails deep dominance (+0.375 against the ≥ 0.40 threshold) at its final layer; the conjunctive outcome is ambiguous and is reported as such (appendix B.1).

Architecture	(H, d)	n	Regime	Gauge	Residual	Crossover	Inertness
ViT-B/16 (anchor)	(12, 768)	5	sup., finetuned	yes	12/12	layer 5 (5/5)	bound
ViT-L/16	(16, 1024)	3	sup., finetuned	yes	24/24	layers 8–11 (3/3)	bound
DINOv2 (ViT-L/14)	(16, 1024)	1	SSL, frozen	yes	24/24	does not transfer	n/a

Metrics. Unsigned minimum angular separation between the representative directions $\mathbf{v}_1(\mathbf{W}_O)$, $\theta_{\min}$, measured per layer—the minimum over pairs within that layer—and reported as the mean across layers; angular redundancy between heads—mean $|\cos|$ over pairs of those directions—; functional similarity s_{func} —cosine between average attention maps—; top-1 accuracy. All are reported with mean and between-seed *band*—throughout this work, “band” denotes the between-seed standard deviation, and that of the base variant is the noise floor against which any effect is contrasted—, and each run’s final values are extracted as the average of the last five converged epochs—the same window for every variant and metric— so that every cell is traceable to its `run.csv`.

Table 3: Fine-tuning protocol, common to base, soft and hard except for the probe term. Source: the repository’s `configs/` and the training loop.

Optimiser	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Base learning rate and schedule	10^{-4} ; cosine over 30 epochs, no warm-up
<i>Weight decay</i>	0.05, decoupled
Batch size	128 (ViT-B, DINOv2); 64 (ViT-L)
Loss	cross-entropy with label smoothing 0.1
Epochs and read-out window	30; average of the last 5 with $\text{ce} > 0$
Augmentation	random resized crop to 224 and horizontal flip; validation: resize to 256 and centre crop
Precision	fp32 in training; fp64 in the gauge-flip
Starting checkpoint (timm 1.0.22)	<code>vit_base_patch16_224</code> <code>vit_large_patch16_224</code> <code>vit_large_patch14_dinov2.lvd142m</code>
Frozen column	fixed backbone; linear head at 10^{-3}
Probe: λ and threshold	10^{-3} ; $\arccos(1/(H - 1))$
Hard variant	reprojection onto the simplex after every step
Seeds	ViT-B {42, ..., 46}; ViT-L {42, 43, 44}

Frozen probing set. The functional measures share a single validation subset, sampled once with a fixed seed (10 images per class, 1000 in total, stratified), so that sampling noise does not contaminate the per-layer comparisons.

6.1 Hypotheses and tests, fixed before the data

This work makes one claim—the dominant direction of the output projection is not a functionally identifiable representation of the head—and spreads the confirmatory discipline across its measured supports. The non-identifiability piece (section 3) is closed-form and does not enter this discipline. The three empirical contrasts are pre-specified here, with their metrics and directions fixed.

H1 (inertness). The probe on $\mathbf{v}_1(\mathbf{W}_O)$ (eq. (7)) raises the angular separation up to the simplex threshold, $\theta_{\min} \rightarrow \theta^*$, but the aggregate functional similarity does not drop below the base level:

$$\Delta s_{\text{func}} = s_{\text{func}}^{\lambda} - s_{\text{func}}^{\text{base}} \approx 0, \quad (9)$$

at the fixed intensity ($\lambda = 10^{-3}$). The prediction is the *absence* of an effect, not its presence, and it is judged against the between-seed noise floor: inertness holds if $|\Delta s_{\text{func}}|$ does not exceed the base band while $\theta_{\min}$ saturates. The criterion is a pre-specified practical equivalence criterion, not a statistical equivalence test. H1 requires training with the probe and is tested on the two finetuned columns (ViT-B and ViT-L). The frozen column does not participate by construction.

H2 (operational non-identifiability). Under the gauge-flip of the value-output sector (eq. (1)) applied without retraining, the drift of $\mathbf{v}_1(\mathbf{W}_O)$ grows with the gauge magnitude $\delta\mathbf{R}$ until it saturates, while the output and the OV-circuit signature remain at the numerical floor. The test is the monotonicity of the weight drift versus its constancy in the circuit, over the strength grid.

H3 (relocation). The per-layer correlation between s_{func} and the $Q \cdot K$ interaction overtakes that of the $\mathbf{W}_O$ subspace at an interior layer, with $Q \cdot K$ going from near zero to clearly positive at that layer—the abrupt jump, not a crossing by margin—and growing monotonically towards the deep layers. The crossover is reproducible *across architectures*, by seed within each one and across the three scales and regimes, and is indifferent to the probe. In the anchor the crossover falls at layer 5. For the new columns only its *existence and uniqueness* are pre-specified—an abrupt jump at an interior layer, with monotonic growth towards the deep layers—not its index: a 24-layer column need not cross “at layer 5”. The response signature is, moreover, verified to be head-specific against a **permutation null** (appendix B.2).

All three are read with their own band and with each architecture’s own n . The confirmatory discipline rests on the *reproducibility of the sign and of the crossover*, not on inferential power tests: H1—absence of movement in s_{func} —is required to be reproducible in the two finetuned columns, and H3—crossover towards $Q \cdot K$ —in all three, so that “five out of five in a single one” is elevated to its reproduction across two scales and two regimes, a firmer criterion than the power that two fewer seeds sacrifice in the new architectures. The prior exploratory smoke run fixed the noise band. The values are confirmed here on clean ImageNet-100 with the frozen set.

Outcome of the confirmatory discipline. H2 and the permutation null reproduce across the three columns. H1 reproduces in the two finetuned ones, with the sign of the shift inverted between them. H3 does not reproduce under the pre-specified terms: it holds in the anchor, remains ambiguous in ViT-L by one of the three thresholds, and does not hold in the frozen column. The criterion is applied exactly as it was fixed and the outcome is reported without readjusting it (appendix B.1).

7 Verification

The verification runs from the statement to its consequence: the orbit of proposition 1, measured pair by pair on the real weights; the pruning decision under the gauge, in vision and in language; the gauge training itself selects, with none injected; and the two regimes that train, the probe up to the threshold and the hard imposition. Results close over five seeds on a ViT-B/16 finetuned on ImageNet-100 and are contrasted on ViT-L/16 and a frozen DINOv2. The language column (Pythia-410M) enters without training, by the same closed form. Two correlational diagnostics—the relocation towards $Q \cdot K$ and the residual carrier—are subordinated to appendix B.

7.1 The orbit, measured pair by pair

A head’s contribution to the residual stream is invariant under a gauge freedom of the value space: for any $\mathbf{R} \in GL(d_h)$, the substitution $\mathbf{W}_v \rightarrow \mathbf{W}_v \mathbf{R}$, $\mathbf{b}_v \rightarrow \mathbf{b}_v \mathbf{R}$, $\mathbf{W}_O \rightarrow \mathbf{R}^{-1} \mathbf{W}_O$ leaves the function intact, because $\mathbf{W}_v \mathbf{W}_O$ —the OV circuit, the sector’s gauge invariant—does not change. The dominant direction $\mathbf{v}_1(\mathbf{W}_O)$ —its first right singular vector, the writing direction into the residual stream—, in contrast, shifts with $\mathbf{R}$: the gauge’s orthogonal subgroup preserves it, and it is the non-orthogonal part of $\mathbf{R}$, present in every generic transformation, that moves it. Applying that gauge to the finetuned base model, without retraining, we measure what each method reads (forward passes and signatures in fp64): the output does not change—relative error $\leq 2 \times 10^{-13}$ across the 420 layer $\times$ seed $\times$ strength combinations—, but the separation between the $\mathbf{v}_1(\mathbf{W}_O)$ —the quantity redundancy-based pruning compares—shifts, while the OV-circuit signature—its

first singular vector, by exact SVD— stays twelve orders of magnitude below ($\leq 4.2 \times 10^{-13}$), at the double-precision floor.

The shift in $\mathbf{v}_1(\mathbf{W}_O)$ is not the accident of one specific transformation: it grows with the gauge magnitude $\delta_{\mathbf{R}}$ and saturates towards orthogonalisation —from 0.05 with an $\mathbf{R}$ barely different from a scalar multiple of the identity, it climbs through 0.35 and 0.66, and settles at ~ 0.83 (below the 0.90 of uniform chance in its row space: saturation preserves residual structure) from $\delta_{\mathbf{R}} \approx 1$ onward—, while the OV circuit’s stays at $\sim 3 \times 10^{-13}$ whatever the strength (table 4). It is not “one gauge moves it”, but “every generic gauge —outside a null-measure subvariety containing $\{c\mathbf{Q}\}$ and the pointwise stabiliser of each $\mathbf{W}_O$ — moves it, and none moves the circuit”. This is the closed-form demonstration that reading or reorganising $\mathbf{v}_1(\mathbf{W}_O)$ —redundancy-based pruning, dominant-direction interpretation, routing by representative vector— operates on gauge noise. And it makes the inertness of the regulariser in section 7.4 —which optimises that same non-identifiable object— plausible, without proving it: that an intervention on it does not move functional similarity through any indirect channel is an empirical fact, measured separately, not a consequence of the gauge.

Table 4: Premise, value-output gauge-flip on the base model, without retraining and in fp64 (mean $\pm$ standard deviation over the 60 combinations of 5 seeds $\times$ 12 layers per strength). Drift is the mean $1 - |\cos|$ between the signature before and after the gauge; $\delta_{\mathbf{R}}$, is not a norm: it measures the mean deviation of $\mathbf{R}$ from its best scalar multiple of the identity, and is the strength dial of the gauge (it grows as $\mathbf{R}$ moves away from the scalar family); the second column is the distance of eq. (2) to the harmless class, the one the theory names. The only class that leaves $\mathbf{v}_1$ invariant for every $\mathbf{W}_O$ is the conformal orthogonal one $\{c\mathbf{Q}\}$, of measure zero in $GL(d_h)$ and never reached by the sampling. The drift of $\mathbf{v}_1(\mathbf{W}_O)$ grows with gauge strength. That of the OV circuit does not: it stays at the double-precision floor, twelve orders of magnitude below, constant and independent of strength. The last two columns materialise at each strength the pruning decision by $\mathbf{v}_1(\mathbf{W}_O)$ —most redundant pair and top-3 set— over those same 60 combinations: the fraction in which the pair changes with respect to the gauge-free decision, and the mean top-3 overlap. The protocol is this table’s —one $\mathbf{R}$ per layer and strength—, unlike table 5, which aggregates five replicas per layer at $\delta_{\mathbf{R}} \approx 1$ and reports 92.7 % there. Here that strength saturates at 1.00, 60 out of 60.

$\delta_{\mathbf{R}}$	$d(\mathbf{R})$ (eq. 2)	Output error	Drift $\mathbf{v}_1(\mathbf{W}_O)$	OV-circuit drift	Pair changes	Top-3 overlap
0.06	0.044 ± 0.000	1.1×10^{-15}	0.05 ± 0.04	2.9×10^{-13}	0.03 ± 0.08	0.91 ± 0.04
0.13	0.088 ± 0.000	1.1×10^{-15}	0.15 ± 0.09	2.9×10^{-13}	0.13 ± 0.05	0.91 ± 0.03
0.25	0.172 ± 0.001	1.1×10^{-15}	0.35 ± 0.16	2.9×10^{-13}	0.42 ± 0.00	0.76 ± 0.03
0.50	0.313 ± 0.001	1.1×10^{-15}	0.66 ± 0.15	2.9×10^{-13}	0.58 ± 0.06	0.49 ± 0.06
1.00	0.471 ± 0.002	8.0×10^{-15}	0.83 ± 0.04	2.9×10^{-13}	1.00 ± 0.00	0.33 ± 0.06
2.00	0.525 ± 0.002	8.5×10^{-15}	0.83 ± 0.05	2.9×10^{-13}	0.93 ± 0.04	0.28 ± 0.07
4.05	0.531 ± 0.002	2.9×10^{-14}	0.82 ± 0.07	2.9×10^{-13}	0.95 ± 0.05	0.28 ± 0.05

The same grid answers under which gauge the decision breaks, not only how far the direction drifts. With the pruning the practice performs materialised at each strength, the most redundant pair changes in 3.3 % of the 60 combinations at $\delta_{\mathbf{R}} = 0.06$ and already in 13.3 % at $\delta_{\mathbf{R}} = 0.13$. The pre-registration of this curve expected stability —pair changed in under 5 %— up to that strength; the 13.3 % exceeds it and the expectation is not met. It is reported as such. The 40 % threshold is crossed at $\delta_{\mathbf{R}} = 0.25$ (41.7 %) and saturation arrives at $\delta_{\mathbf{R}} = 1$, where none of the 60 keeps its pair. The reading is not that every gauge perturbation breaks the decision: the conformal orthogonal class, which leaves $\mathbf{v}_1$ exactly invariant, has a neighbourhood, and at the weak end of the grid the decision survives —continuity of singular vectors with a positive gap—, even though the certified radius over the orbit is null (proposition 1). It is that the neighbourhood is narrow —below $\delta_{\mathbf{R}} \approx 0.1$, with $\mathbf{R}$ within a tenth of its deviation from the scalar family— and that at full strength the decision keeps nothing. At the two largest strengths the fraction eases to 0.93 and 0.95 without returning to 1. With 66 pairs per layer, the chance-coincidence floor leaves of the

order of one stable combination out of 60, and what is observed sits above that floor, close to it. It is reported uncorrected.

proposition 1 is further certified pair by pair on the real weights (appendix A): across the five seeds, the orbit’s floor —the first principal angle between row spaces— drops as low as 14.7° depending on the pair, the simplex threshold $\theta^* = 84.78^\circ$ falls within the attainable range of the 3960 pairs measured ($66 \text{ pairs} \times 12 \text{ layers} \times 5 \text{ seeds}$), and a descent restricted to the row spaces finds, across the 60 layer $\times$ seed combinations, a joint configuration of the twelve directions with $\theta_{\min} \geq \theta^*$ (worst case 84.79°). *Total* compliance of the regulariser therefore lies within the gauge orbit: there is a path of exactly zero functional cost towards it. Every observed angle between dominant directions is a point on a range tens of degrees wide that the gauge sweeps with the function fixed. The specific value is a choice of gauge, not of function. How much of that choice training fixes on its own, and whether it suffices, is measured in section 7.3.

7.2 The pruning decision under gauge: vision and language

The drift is not abstract: the decision the practice would make changes under the gauge. Materialising the two typical decisions —the most redundant pair of heads and the top-3 pruning set by mean redundancy— on every layer and seed, and applying five gauges per layer at saturated strength ($\delta_{\mathbf{R}} \approx 1$), the decision by $\mathbf{v}_1(\mathbf{W}_O)$ changes pair, under generic reparametrisation, in 92.7% of cases and retains on average 0.38 of the top-3 (per layer, between 0.24 and 0.69). The same decision taken with the gauge-invariant response signature never changes pair and retains the full top-3 (table 5). Two functionally identical models, under the weight criterion, prune different sets of heads.

The instability is not a property of vision or of finetuning: it reproduces, by the same closed form and without training anything, on a pretrained language transformer. On Pythia-410M (GPT-NeoX; Biderman et al., 2023) (24 layers, 16 heads, pure multi-head attention) the same value-output gauge, at an equivalent pruning budget ($k=4$ of 16, the same relative 25% as the anchor’s top-3 of 12), changes the most redundant pair in 90.0% of cases and retains 0.379 of the pruning set (table 5) —an overlap comparable to the 0.378 of the anchor: the same closed form breaks the decision to a comparable degree across two modalities and two architectures—. The response signature requires forward passes with a probe and is not ported to this column. The comparison is limited to the weight criterion.

Table 5: Decision under gauge, without retraining. ViT-B: 5 seeds $\times$ 12 layers $\times$ 5 gauges at $\delta_{\mathbf{R}} \approx 1$ ($d(\mathbf{R})$ of eq. (2) between 0.468 and 0.476), top-3 of 12 (25%). Pythia-410M: 24 layers $\times$ 5 gauges at the same saturated strength, top-4 of 16 (same relative 25%; frozen column, a single checkpoint). The response signature is not measured on the language column (it requires forward passes with a probe, out of scope). Fraction of cases in which the most redundant pair changes and mean overlap of the pruning set with the no-gauge decision, by criterion. Thresholds fixed before the data: weights unstable if the pair changes in $\geq 40\%$ of cases or the mean overlap < 0.67 ; signature stable if the overlap ≥ 0.95 and the pair changes $< 5\%$. The signature’s declared stability is against the gauge. Its sensitivity to probe sampling is bounded and measured in lemma 2.

Architecture	Criterion	Pair changes (%)	Top- k overlap
ViT-B/16 (anchor, $k=3$)	$\mathbf{v}_1(\mathbf{W}_O)$ (weights)	92.7	0.378
ViT-B/16 (anchor, $k=3$)	Signature $\mathbf{v}_1(\mathbf{C}_h^P)$	0.0	1.000
Pythia-410M (language, $k=4$)	$\mathbf{v}_1(\mathbf{W}_O)$ (weights)	90.0	0.379
Pythia-410M (language, $k=4$)	Signature $\mathbf{v}_1(\mathbf{C}_h^P)$	not measured	not measured

7.3 The gauge training selects, and how far it reaches

The two measurements in this subsection were not part of the original design (section 6). They were pre-registered during the revision of this work, with statistic, aggregation and thresholds

fixed in writing before anything was run, in order to answer in advance the strongest objection against proposition 1. The objection is this. *Weight decay* —decoupled in the optimiser used (table 3)— exerts pressure towards factors of smaller Frobenius norm, and the minimum of that norm over the orbit at fixed circuit is the balanced factorisation (lemma 1), unique up to the orthogonal subgroup —under which, by section 3, $\mathbf{v}_1(\mathbf{W}_O)$ is invariant—. Under exact balance one further has $\mathbf{v}_1(\mathbf{W}_O) = \mathbf{v}_1(\mathbf{W}_v \mathbf{W}_O)$: the proxy would be reading the identifiable object, and non-identifiability, true as a worst-case impossibility, would be harmless in practice. Were training to select a nearly canonical gauge, proposition 1 would not bite where it matters. That is a measurable hypothesis, and it is measured.

One object needs fixing first, one that was not needed until now. The *static invariant* $\mathbf{v}_1(\mathbf{W}_v^{(h)} \mathbf{W}_O^{(h)})$ is the dominant direction of the OV circuit, computed on the weights and with no data at all. It is not the response signature $\mathbf{v}_1(\mathbf{C}_h^P)$, which is this work’s instrument and does require data to pass through the model. Both are gauge-invariant, but only the second measures what the head computes. This subsection compares the weight reading against the static invariant, because that is the exact term of the objection: were the gauge canonical, those two would coincide.

Training canonicalises: alignment improves monotonically across the three observed states. The canonicalisation spoken of here is the approach to the point where the proxy would coincide with the invariant. Per head, layer and seed we measure

$$\alpha_h = \angle(\mathbf{v}_1(\mathbf{W}_O^{(h)}), \mathbf{v}_1(\mathbf{W}_v^{(h)} \mathbf{W}_O^{(h)})),$$

which is 0° under exact balance and over the whole aligned family that follows from lemma 1, the QR canonical form included. On the fine-tuned checkpoints the median is $35.20 \pm 2.11^\circ$ on ViT-B and $18.14 \pm 0.24^\circ$ on ViT-L, far from the floor of two random directions, which depends on the residual dimension (88.5° median and 85.9° fifth percentile in $\mathbb{R}^{768}$; 88.8° and 86.4° in $\mathbb{R}^{1024}$). The objection therefore has an empirical basis: the gauge that training selects is no adversary—it moves towards the canonical point. A diagnostic outside the pre-registration places it on its trajectory (table 6): α decreases across the three measured states, from $\sim 65^\circ$ at initialisation to $\sim 40^\circ$ after pre-training and to 35° or 18° after fine-tuning. At ViT-L scale the two pre-training regimes canonicalise to within a degree of each other — 38.25° the supervised one and 39.00° the self-supervised one of DINOv2, which, being frozen, stays there—, and it is fine-tuning that covers the rest down to 18° : canonicalisation by pre-training denotes a property of descent more than of the regime. That of fine-tuning is measured only in the supervised regime. The robustness reading, pre-registered, rules out a numerical artefact. Restricted to heads with a clear spectral gap ($\sigma_1/\sigma_2 \geq 1.05$, somewhat more than half), the median *drops* to $30.07 \pm 1.70^\circ$ and $13.94 \pm 0.65^\circ$: where $\mathbf{v}_1$ is better defined, it aligns more.

But canonicalisation does not reach the decision. The question that matters is whether those degrees change what is decided, not how many remain. On the same checkpoint and without injecting any gauge, the two decisions of table 5 —most redundant pair and the 25 % pruning set by mean redundancy— are materialised under two criteria: the weight reading $\mathbf{v}_1(\mathbf{W}_O)$ and the static invariant. The resulting agreement (table 7) is well above chance: the top- k overlap is 0.61 and 0.64 against the 0.25 of choosing blindly, and the pair coincides in 63 % and 56 % of layers against the 1.5 % and 0.8 % of chance. Reading $\mathbf{v}_1(\mathbf{W}_O)$ in the natural gauge is not reading noise. And yet the agreement stays below the bar this same work set for calling a decision stable —overlap ≥ 0.67 and pair constant in at least 60 % of cases—: on ViT-B it fails on the overlap (0.61), with the pair at the limit ($63 \pm 13\%$), and on ViT-L it fails on both. The most redundant pair differs between the two criteria in 37 % and 44 % of layers. The residual misalignment of 18 to 35° suffices to corrupt the decision without touching anything. The pre-registered descriptor confirms it in the expected direction: the per-layer correlation between mean α and overlap is -0.52 and -0.38 .

Table 6: Alignment between the weight reading and the static invariant, $\alpha = \angle(\mathbf{v}_1(\mathbf{W}_O), \mathbf{v}_1(\mathbf{W}_v \mathbf{W}_O))$, median over heads $\times$ layers (mean $\pm$ deviation across seeds; init and pre-trained are a single realisation and carry no band). The fifth column restricts the median to heads with a clear spectral gap, on each row’s measured state —fine-tuned in the two supervised ones, frozen in DINOv2—. Two random directions give 88.5° in $\mathbb{R}^{768}$ and 88.8° in $\mathbb{R}^{1024}$. Fine-tuning brings the reading closer to the identifiable object without taking it there. The primary statistic is α and not the spectral imbalance δ , for a reason the control itself exposes: at initialisation δ_{shape} is *smaller* (0.006) than after training (0.019) while α is *larger* (65° against 40°). Two independent Gaussian matrices have nearly identical spectra and wholly unrelated bases: δ measures the balance of the spectra, but the balance that makes the proxy coincide with the invariant further requires the bases to match, and only α sees that. A diagnostic, outside the pre-registration.

Column	Init	Pre-tr.	Fine-tuned	$\sigma_1/\sigma_2 \geq 1.05$	δ_{shape}
ViT-B/16 ($n=5$)	64.61	42.91	35.20 ± 2.11	30.07 ± 1.70	0.020
ViT-L/16 ($n=3$)	64.83	38.25	18.14 ± 0.24	13.94 ± 0.65	0.017
DINOv2 (ViT-L/14, $n=1$)	64.06	39.00	—	29.09	0.021

Table 7: Agreement between the decision by $\mathbf{v}_1(\mathbf{W}_O)$ and the decision by the static invariant, in the gauge training left and with none injected (mean $\pm$ deviation across seeds over layers $\times$ seeds). Thresholds inherited from table 5, fixed before this measurement existed. It complements table 5: there the decision breaks under generic reparametrisation. Here it does not even coincide without reparametrising.

Column	Top- k overlap	Chance	Pair coincides	Chance
ViT-B/16 ($k=3$ of 12)	0.611 ± 0.020	0.25	$63.3 \pm 12.6 \%$	1.5 %
ViT-L/16 ($k=4$ of 16)	0.639 ± 0.033	0.25	$55.6 \pm 6.4 \%$	0.8 %

Two clarifications on the reach of these two tables. First: they measure *agreement between criteria*, not pruning quality; which heads to prune is the question of table 9, where no similarity —of weights or of signature— beats chance, and the two questions are not to be mixed. Second: that the agreement exceeds chance does not authorise reading $\mathbf{v}_1(\mathbf{W}_O)$ as a usable approximation of the invariant. An overlap of 0.61 below the work’s own bar supports no recommendation of use. It supports the opposite, that this is a reading whose reliability depends on a gauge not chosen on purpose. The functional counterpart of that reading —what happens when one *intervenes* on that object instead of reading it— is the subject of section 7.4, and there inertia is measured on the model’s output as well.

7.4 The regulariser on $\mathbf{W}_O$ separates at no detectable functional cost

The weight carrier is the representative direction $\mathbf{v}_1(\mathbf{W}_O^{(h)})$, the output projection’s first right singular vector, and the regulariser is that of eq. (7) on that carrier. Finetuning alone separates the heads only partially: $\theta_{\min}$ stalls at 51.96° , well below the simplex threshold ($\theta^* = 84.78^\circ$). The stall does not indicate a packing obstruction: with $K=12 \leq d=768$ the unsigned optimum is the orthogonal frame at 90° , with 756 dimensions of slack, and the admissible interval $|\cos| \leq 1/11$ is enormous. It is a non-obstruction certificate, and the 51.96° turns from raw data into directed evidence. The soft regulariser drives that separation to 85.01° —it clears the threshold and keeps drifting towards the orthogonal frame— and reduces angular redundancy from 0.172 to 0.034, with a 0.10 pp accuracy drop that falls *within* the base’s between-seed top-1 band (0.19 pp): that, not a nominal zero, is what “at no cost” means. The hard variant, which reprojects onto the simplex after every step, is pinned to it (84.77°) —below the soft one, which clears it without any code at all: the hinge drifts freely towards the orthogonal frame while the reprojection fixes it to a suboptimal configuration— and pays 4.40 pp —uniform across seeds (range 0.86 pp), 23 times the band— and destabilises accuracy within each run (intra-run deviation 0.0022 versus 0.0013 for

base). The soft variant not only matches the hard one in separation, it beats it in de-correlation. Table 8 records the final state.

The pattern repeats in the deep column: the hard variant pays 5.6 pp —uniform across seeds— and shifts $s_{\text{func}} +0.031$, 31.8 times ViT-L’s base band, while the soft variant reaches the same θ_{min} at no measurable cost. The hard variant’s shift keeps its sign and order of magnitude across architectures. The soft one’s inverts them and stays in the noise.

On s_{func} , the heart of H1: the shift between variants is of the order of the between-seed spread ($\Delta = +0.0013$, 95 % CI $[-0.0052, +0.0077]$, which includes zero; Student’s t with 4 degrees of freedom on the seed-paired differences), with no detectable separation from zero across five replicates, with a possible sign bias smaller than that spread. The result is a *bound*, not a null. Its sensitivity analysis: the full between-seed band of base s_{func} is 0.0041, inflated by two atypical runs (one high, one low, of opposite signs). The core without them is 0.0017. The effect (0.0013) falls within both measures, and the seed-paired reading agrees with the reading against the base mean except for the two atypical seeds —the dual pairing that detects them—, so that effect, between-seed noise, and run noise are of the same size and none separates out. The hard variant, by contrast, does *not* leave s_{func} still: it shifts it $+0.052$ upward —12.8 times the band, uniform across the five seeds—, that is, attention maps become *more* similar across heads while the weight geometry ends up maximally separated. The asymmetry is itself a result: two mechanisms —gradient descent on the penalty for one, forced reprojection without backpropagation for the other— reach a comparable minimum separation (0.23° apart) with different global configurations —redundancy 0.034 against 0.091—, and one produces no detectable cost while the other drags the function. The inertness H1 establishes is that of the *gradient probe* at the fixed intensity, not that of any manipulation of that geometry: forcing the weights off the optimiser’s trajectory costs accuracy and moves functional similarity. Numerical detail of the soft probe: the SVD gradient degrades near convergence. θ_{min} saturates around epoch 6 but s_{func} drifts up to epoch ~ 25 and stabilises afterward (epoch-to-epoch deviation 0.0007), which is why the metrics are extracted from the converged window (section 6).

Table 8: Premise, part I: ablation of the angular regulariser on the geometry of $\mathbf{W}_O$ (clean ImageNet-100, 30 epochs, mean $\pm$ standard deviation across 5 seeds; per-run values averaged over the converged window). θ_{min} is the unsigned minimum angular separation between representative directions, per layer and reported as the mean across layers (threshold 84.78°); redundancy is angular (mean $|\cos|$ between directions). The soft variant separates at a cost within the between-seed band ($0.10 < 0.19$ pp) and a shift in s_{func} bounded by that same spread. The hard variant pins the code, paying 4.40 pp and moving $s_{\text{func}} +0.052$ upward: two mechanisms, the same minimum separation, different global configurations, and only one pays.

Configuration	$\theta_{\text{min}} \uparrow$	Redundancy $\downarrow$	Top-1 $\uparrow$	s_{func}
Base ($\lambda_A = 0$)	$51.96^\circ \pm 0.26$	0.1724 ± 0.0030	0.9284 ± 0.0019	0.6945 ± 0.0041
Soft ($\lambda_A = 10^{-3}$)	$85.01^\circ \pm 0.08$	0.0338 ± 0.0009	0.9274 ± 0.0017	0.6958 ± 0.0028
Hard (reprojection)	$84.77^\circ \pm 0.00$	0.0909 ± 0.0000	0.8844 ± 0.0021	0.7466 ± 0.0049

Inertia, measured at the output. The functional similarity s_{func} compares attention maps, and $\mathbf{A}_h = \text{softmax}(\mathbf{Q}\mathbf{K}^\top)$ does not contain $\mathbf{W}_O$ by any direct route: an intervention on $\mathbf{W}_O$ could stay inert there by construction rather than by substance. The objection is closed by measuring where the model stakes its function. On the frozen probe set we compare the Kullback-Leibler divergence between the output distributions of the base run and the soft one of the *same* seed, against the range that changing the seed produces within the base itself, and with the hard variant as a positive control. On ViT-B the paired base-soft divergence is 0.118 nats of per-image mean, below the minimum of the between-seed range ($[0.121, 0.157]$, $n=20$): the soft probe perturbs the

output less than changing the training seed does. In three of the five seeds that holds individually. On ViT-L the paired divergence is 0.185 against a range of [0.169, 0.201]: within the noise, at its centre, not below it. The hard variant leaves that range in both columns (0.253 and 0.308), so the instrument does detect a damaging intervention when there is one. The divergence has a long tail, and both figures are therefore reported: the per-image median is 0.0025 against 0.0038 of the range and 0.0115 of the control on ViT-B, an order of magnitude below the mean in all three cases. The drift of the response signature from base to soft likewise stays within its between-seed range (0.194 against a ceiling of 0.223 on ViT-B; 0.359 against 0.394 on ViT-L). The inertia of the soft probe is therefore no artefact of looking only at attention maps: it holds in the logits, with the soft/hard asymmetry reproduced on a channel s_{func} does not see.

Inertness is established above at the fixed intensity $\lambda = 10^{-3}$: θ_{min} saturates without s_{func} dropping below the base level. Characterising the dependence on λ over a wider range remains as an extension, and would require the same reading apparatus—five seeds per intensity, against the between-seed noise floor—that the support at $\lambda = 10^{-3}$ already consumes. The relevant caveat: at much higher intensity the probe could stop being an inert perturbation and start deforming the representation, so indifference is not asserted beyond the measured intensity.

8 Discussion and limitations

The three arms, read together. Gauge, probe, and hard variant are not three loose results: they characterise, in three intervention regimes on the value-output sector that differ by mechanism, the coupling between that geometry and the function. The freest intervention—the gauge, which does not touch the function by construction—moves the geometry to the maximum at no cost; the intermediate intervention—the probe, which does train—also moves it at no cost detectable at the measured strength; only the most aggressive one—the hard variant, which imposes the simplex at every step—pays in function. That the damage appears exactly where the intervention stopped being a gauge and started being an active constraint on the optimiser’s trajectory is itself the datum distinguishing “reparametrisation” from “manipulation”, not a post-hoc reading. The asymmetry connects with the relocation: if the function does not live in $\mathbf{v}_1(\mathbf{W}_O)$, and the organisation of attention does correlate with $Q \cdot K$ in the fine-tuned regime, the constructive question—which component governs the function, if not that direction—remains open, not answered. Component-wise attribution over $Q \cdot K$ is the natural continuation (section 7.4 and appendix B.1).

What the result claims and what it does not. The claim is negative and bounded: the geometry of the output projection— $\mathbf{v}_1(\mathbf{W}_O)$, the direction that redundancy-based pruning and interpretation by dominant direction read—is not a functionally identifiable representation of the head. It is not identifiable under the value-output sector’s gauge, separating it produces no detectable change in functional similarity, and the organisation of attention resides in another substrate. The work does not claim the opposite of that: it does not hold that an actionable weight carrier exists, nor that $Q \cdot K$ is the only substrate of the function—that reading is correlational—, nor that the response signature exhausts the available diversity. Intervening constructively on $Q \cdot K$, or on the response signature, is the natural continuation, not a result of this work.

Implications for pruning and interpretation. Part of head pruning decides redundancy by comparing representative directions of $\mathbf{W}_O$, and part of mechanistic interpretation reads a head’s role from its dominant weight direction. Section 7.1 shows that those comparisons operate on a non-identifiable quantity: two functionally identical models exhibit different separations between their $\mathbf{v}_1(\mathbf{W}_O)$ ’s depending on the gauge, so that angular redundancy measured on weights is, in part, a reparametrisation artefact, and the decision it produces does in fact change under gauge (table 5). The gauge-invariant response signature (section 4) offers the quantity those tasks need

as a *measure*: head diversity anchored to the identifiable circuit, which does not move under reparametrisation and is head-specific (appendix B.2). With a measured caveat: stability is not a selection criterion —no similarity, of gauge or of function, beats the random floor as a pruning rule at a fixed budget (table 9). Deciding *what* to prune still calls for importance criteria such as second-order ones (Yang et al., 2023; Uddin et al., 2025), on which this analysis takes no position—.

Specialisation is not captured by the isolated dominant direction of W_O . If heads specialise functionally (Michel et al., 2019; Voita et al., 2019), the results suggest that specialisation is read in computed behaviour and not in the dominant direction of the output weights. Whether another reading of the weights —the circuit, the subspace, $Q \cdot K$ — carries it is an open question. A neighbouring line of work that regularises diversity over input gradients —a signal tied to behaviour— alters the function and improves OOD generalisation (Nicolicioiu et al., 2023), whereas intervening on the weight geometry does not move it detectably: the contrast places the effect in the functional signal, not in the static summary of the weights.

Limits. The non-identifiability is closed-form and does not depend on the architecture. Inertness and relocation live within it, and are for that reason contrasted across three columns (table 2): two scales and depths at constant per-head width —ViT-B and ViT-L— and two training regimes —supervised and self-supervised, the latter with frozen DINOv2—. Gauge invariance, the broken decision and the static diagnostics reproduce in the self-supervised column. H1 is not applicable without training and H3 does not transfer. The evidence on inertness is restricted to the two finetuned columns, and relocation to the supervised finetuned regime. The declared boundary is non-global attention: in Swin and window attention the value-output non-identifiability would indeed transfer —the value-output gauge is structural—, but the shifted-window local map, the relative position bias in the logits, and the stage-wise hierarchical depth require re-deriving s_{func} and the very notion of depth before measuring inertness or relocation. It remains an extension, not a hole in rigor. The signature’s exact stability is against the gauge; against probe sampling it is a finite matter, and pair coincidence across halves grows with probe size without saturating over the measured grid —from 10.4 to 11.4 over 12 layers between $P=250$ and $P=5000$ (lemma 2)—: whoever uses the signature at a given size inherits that sensitivity, and the curve says how much of it is left. The reading of the relocation towards $Q \cdot K$ is correlational —Spearman between s_{func} and geometric candidates per layer—, not causal, and the choice of the first singular vector of C_h^P as the signature, against other summaries of the response, is a design decision whose contrast with patch-centring bounds its sensitivity without exhausting it. The deep-dominance threshold inherited the anchor’s assumption that the last third of layers is homogeneous. ViT-L breaks it at its final layer, and the assumption is declared as a limit of the criterion, not corrected after the fact.

8.1 Similarity is not a pruning criterion; stability is what the instrument offers

A minimal benchmark bounds what the instrument buys and what it does not. With the same budget ($k=2$ heads per layer, 24 of 144) and the same pruning mechanism —zeroing out the head’s columns in the output projection—, the most redundant set is pruned according to each criterion and the drop in top-1 is measured on the full validation set (table 9). No similarity criterion beats the random floor: selection by weights falls within the random null of a hundred draws per seed (3.13 ± 0.54 pp against a mean of 2.89 pp, with 5th and 95th percentiles at 1.94 and 3.96; percentile 62 of the pooled null and below its 5th percentile in no seed) —consistent with the fact that choosing by $v_1(W_O)$ is choosing by gauge noise—, and selection by signature does *more* damage than chance (4.26 ± 0.29 pp, percentile 97 of the null and above its 95th percentile in four of five seeds, the pre-registered criterion). The diagnostic explains why: both criteria select heads with a large contribution to the residual ($2.7\times$ the mean energy of the remaining ones), and the ones

the signature groups are also individually more costly (pruning half the pair at a time, on seed 42, the sum of the halves is already 3.72 against 2.90 pp for weights, with similar superadditivity in both, +0.8/+0.9 pp). Similarity —of gauge or functional— is not equivalent to dispensable redundancy: heads with similar signatures are central, not surplus. The applied statement, for practice, joins the two certificates: the decision by $\mathbf{v}_1(\mathbf{W}_O)$ admits no positive certified radius for any admissible threshold —the gauge fabricates it at will (proposition 1 and table 5)— and the decision by signature is invariant. Neither similarity, however, selects what to prune better than chance at this budget. The instrument’s advantage is the *stable decision* under reparametrisation, not the selection; what to prune calls for importance criteria, outside the scope of this analysis.

Table 9: Pruning by criterion, without retraining (5 base seeds; $k=2$ heads per layer, 24 of 144; mean $\pm$ between-seed band). The random null is a hundred draws per seed, 500 in total; the mean of the three original draws, 3.25 pp, is reproduced at 3.26 (regression row: the original 3.25). The pruning mechanism is identical for all criteria. No similarity beats the null —pre-registered criterion: drop below its 5th percentile in at least four of five seeds—, and the signature harms more than chance in four of five. The instrument’s role is decision stability, not selection.

Criterion	Pruned top-1	Drop (pp)	Percentile in the null
$\mathbf{v}_1(\mathbf{W}_O)$ (weights)	0.898	3.13 ± 0.54	62
Signature $\mathbf{v}_1(\mathbf{C}_h^P)$	0.887	4.26 ± 0.29	97
Random null (100 draws)	0.900	2.89 ± 0.18	p5 1.94, p95 3.96
Random, 3 draws (regression)	0.896	3.25 ± 0.19	—

The practical effect of this analysis is protective: it identifies a reading of weight space whose conclusions change under reparametrisations that do not touch the function and replaces it with an invariant one. It introduces no new capabilities and uses only public benchmark data.

9 Conclusion

The dominant direction $\mathbf{v}_1(\mathbf{W}_O)$ —the one that redundancy-based pruning and interpretation by dominant direction read— is not identifiable under the value-output gauge. Its orbit is the full sphere of the row space, and the gauge shifts it —up to 0.83 of mean drift on the anchor and 0.91 on ViT-L— while the output and the OV circuit $\mathbf{W}_v \mathbf{W}_O$ remain at the numerical floor of double precision, an invariance verified identically across the three columns, including the frozen self-supervised one. No similarity threshold on $\mathbf{v}_1(\mathbf{W}_O)$, within the range the row spaces allow, admits a positive certified radius, and every gauge-invariant readout of $\mathbf{W}_O$ alone factors through its row space (corollary 1).

The consequence is not caution: two functionally identical models prune different sets of heads —the most redundant pair changes in 92.7 % of cases, under generic reparametrisation, on the fine-tuned ViT and in more than 90 % on a language transformer with no training at all, and the pruning-set overlap drops to 0.378 on the anchor—, while the same decision taken on the response signature remains intact. The criterion does not choose *badly* what to prune: it is not well defined on functional equivalence classes.

In the gauge training leaves, the weight reading moves towards the identifiable object without the decision coinciding (section 7.3). The quantity being read is the gauge choice that training fixes systematically but insufficiently.

For anyone pruning, interpreting or routing by the geometry of $\mathbf{W}_O$, the response signature $\mathbf{v}_1(\mathbf{C}_h^P)$ —probe-conditioned, invariant under the gauge and head-specific (appendix B.2)— is what takes its place, with the finite-sample sensitivity that lemma 2 bounds and the probe curve measures.

The three intervention regimes on the sector complete the picture as secondary evidence of the decoupling. Separating that direction angularly up to the simplex threshold produces no detectable functional cost at the measured strength: the probe raises $\theta_{\min}$ from 52° to 85° on the anchor and from 63° to 86° on the deep column, and the shift in s_{func} remains on the order of the between-seed dispersion in both, with the sign inverted between them $+0.0013$ and -0.0019 . The same criterion distinguishes the opposite case: the hard variant, which imposes the simplex by construction, shifts s_{func} by $+0.052$ and $+0.031$ —12.8 and 31.8 times the band, same sign in both— at a price of 4.4 and 5.6 points of accuracy. The soft probe’s inertness is not, therefore, an incapacity of the measure. The relocation of attention organisation towards $Q \cdot K$ with depth is bounded to the supervised fine-tuned regime —clean on the anchor, sustained and qualified on ViT-L, absent on the frozen DINOv2—, so that H3, as pre-registered for the three columns, does not hold.

Code and data availability

The code for the angular probe and all measurements, together with the gauge-orbit certification (appendix A), is available in the open repository <https://github.com/mmunozp1/angular-separation-vit>, archived on Zenodo at <https://doi.org/10.5281/zenodo.2163053>. The pre-registration of each experiment was written and committed before execution (B1 and B2 on 26 August 2026; E0, E1 and E2 on 21 September 2026) and is published under `prereg/` in the open repository. The CSV files backing every table and figure of section 7 are published at <https://huggingface.co/datasets/ManPla/angular-separation-vit-results> (<https://doi.org/10.57967/hf/9743>), and the reproduction checkpoints —per seed and architecture— at <https://huggingface.co/ManPla/angular-separation-vit-checkpoints> (<https://doi.org/10.57967/hf/9742>). The interactive demo of the gauge orbit —the drift of the direction and the broken pruning decision, on the real weights— is served at <https://huggingface.co/spaces/ManPla/angular-separation-vit-demo>.

Declaration on the use of AI tools

Claude (Anthropic) was used as a methodological-review assistant during the preparation of this manuscript, and Claude Code as an assistant in writing and running the experimental code. The mathematical statements and their proofs, the experimental designs and their pre-registrations, and the conclusions are the author’s own. Every reference cited was verified against its primary source, and every numerical result reported comes from code the author ran and inspected. The author assumes responsibility for the integrity, accuracy, and originality of all content of this submission.

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A The gauge orbit of $v_1(W_O)$, measured

This appendix numerically certifies proposition 1 on the actual weights of the five ViT-B base seeds, without retraining and on CPU.

Per-pair range. For each layer and each pair of heads (h, h') , the principal angles between the row spaces $\mathcal{S}_h$ and $\mathcal{S}_{h'}$ are computed (SVD of $\mathbf{B}_h^\top \mathbf{B}_{h'}$, with $\mathbf{B}$ the orthonormal row bases). The first principal angle is the floor of the joint orbit —no gauge can bring the dominant directions closer than that— and the ceiling is 90° ($d_h \geq 2$ suffices: two principal vectors of distinct index are orthogonal). Over $66 \text{ pairs} \times 12 \text{ layers} \times 5 \text{ seeds}$: global minimum floor 14.7° , and $\theta^* = 84.78^\circ$ within the range of the 3960 pairs.

Explicit realisation. For one pair per seed, the gauge that carries each $\mathbf{v}_1$ to its target is constructed (the pair of directions with minimum angle between the two subspaces): the drift of the OV circuit stays at $\sim 2 \times 10^{-15}$ —invariance at machine precision— with the dominant directions shifted by tens of degrees.

Simultaneous certificate. The per-pair certificate is not enough for the regulariser, which imposes 66 coupled constraints on 12 directions. A descent restricted to the row spaces —free coordinates in the basis of each $\mathcal{S}_h$, penalty on $|\cos| > 1/(H-1)$ — finds, across the $60 \text{ layer} \times \text{seed}$ combinations, a joint configuration with $\theta_{\min} \geq \theta^*$ (worst case 84.79°). Full compliance with the regulariser lies within the joint gauge orbit: a path of exactly zero functional cost to it exists, and the inertness of section 7.4 is “no change happened despite a free path existing”, not “no change was observed”.

B Beyond the value-output sector: two correlational diagnostics

The two diagnostics in this appendix are correlational and did not reproduce across the three columns of the trio (section 6). They are therefore subordinated to the body of the work, which does not depend on them. They bound where the organisation of attention lives when the dominant direction does not carry it.

B.1 The substrate of attention relocates towards $Q \cdot K$ with depth

Beyond the non-identifiability of $\mathbf{v}_1$ (section 7.1), a layer-wise characterisation of the finetuned model shows where the organisation of attention resides and why the geometry of $\mathbf{W}_O$ is inert. For every layer, functional similarity s_{func} —cosine between average attention maps— is correlated with two geometric candidates: the column subspace of $\mathbf{W}_O$ (similarity between the 768×64 per-head spaces via principal angles) and the $Q \cdot K^\top$ interaction (cosine between the pre-softmax logits). Table 10 reports both together with the sole dominant direction $\mathbf{v}_1$. Figure 1 plots them per layer with their between-seed band.

The pattern is an inversion. In the early layers the subspace of $\mathbf{W}_O$ correlates with the function while $Q \cdot K$ does not. From layer 5 onward the order flips and $Q \cdot K$ takes over, growing up to ~ 0.9 in the deep layers. The crossover $r_{Q \cdot K} > r_{\mathbf{W}_O\text{-subsp.}}$ occurs at layer 5 in all five seeds, with $Q \cdot K$ switching on abruptly between layer 4 and 5 ($r_{Q \cdot K}$ from 0.026 to 0.51). The inversion is asymmetric — $Q \cdot K$ goes from ≈ 0 to ≈ 0.9 while $\mathbf{W}_O$ was barely above noise— and independent of the regulariser: under the soft variant the crossover stays at layer 5 and $r_{Q \cdot K}$ is indifferent to the regularisation (maximum $|\Delta| = 0.043$). The dominant direction $\mathbf{v}_1$ that the weight regulariser touches does not participate in the crossover: it is low or negative in ten of the twelve layers, tracing function only in the layer 3–4 transition. That trace is real in the gauge training fixed, but it is not identifiable (section 7.1): another gauge would erase it without touching the function, so it does not contradict the non-identifiability of $\mathbf{v}_1$ —it illustrates it, it is precisely the point that reading via $\mathbf{v}_1$ depends on the gauge—. This is the measured reason for the inertness: the weight regulariser optimises an object that does not govern the organisation of attention.

On ViT-L ($n=3$, 24 layers) the sustained crossover exists in all three seeds —layers 8, 11, and 8, at a relative depth of 33 to 46 %, comparable to the anchor’s 42 %— and it clears two of the three thresholds: the jump at the crossover layer is $+0.403$ (threshold ≥ 0.30) and the margin

+0.366 (threshold ≥ 0.15). Deep dominance lands at +0.375, below the ≥ 0.40 threshold, and the conjunctive outcome is accordingly ambiguous. The diagnosis reveals a localised cause: layer 23 drags down the minimum in all three seeds —+0.43, +0.41, and +0.28— while layers 16 to 22 clear it in every case (+0.61 to +0.63). In the anchor the last third showed no such dip. The final layer is not excluded from the threshold computation, fixed before the data. The threshold is reported as it came out, together with its cause.

Relocation does *not* transfer to the frozen column. On DINOv2 (24 layers, a single checkpoint), the thresholds fixed before the data fail three of four: the sustained crossover only appears from layer 22 onward, with a jump of +0.223 at that point, the margin is +0.009 (threshold ≥ 0.15 , fails), and deep dominance does not exist —minimum $r_{Q\cdot K}$ of the last third -0.059 against the ≥ 0.4 threshold, fails—: in the pure self-supervised backbone, $r_{Q\cdot K}$ does not take off in any depth stretch. H3, as pre-registered for the three columns, does not hold in the frozen regime.

Relocation is thus bounded to the finetuned supervised regime: clean in the anchor, sustained but qualified in the deep column, absent in the frozen one. H3, pre-registered as reproducible across the three columns, does not hold under those terms.

Table 10: Premise, part II: Spearman correlation between s_{func} and two geometric candidates, per layer (base variant, mean across 5 seeds; $r_{Q\cdot K}$ does not depend on k). The subspace is measured at $k = 64$. v_1 is the sole dominant direction. The random baseline of the subspace at $k = 64$ is ≈ 0.25 ; that of $Q\cdot K$, ≈ 0 . The crossover towards $Q\cdot K$ at layer 5 marks where the organisation leaves residual writing and moves to the query-key comparison.

Layer	$r_{\mathbf{W}_O}$ (subsp.)	$r_{\mathbf{W}_O}$ (v_1)	$r_{Q\cdot K}$
0	+0.439	+0.185	−0.104
1	+0.456	+0.075	+0.189
2	+0.503	+0.094	+0.233
3	+0.470	+0.463	+0.212
4	+0.341	+0.223	+0.026
5	+0.150	−0.005	+0.510
6	+0.176	−0.096	+0.296
7	+0.069	+0.038	+0.720
8	+0.058	−0.168	+0.811
9	−0.198	−0.313	+0.907
10	−0.041	−0.340	+0.770
11	+0.089	−0.108	+0.832

B.2 Head identity across probe halves

The head-specificity of the signature is tested here without $\mathbf{v}_1(\mathbf{W}_O)$, with a pre-registered gauge-invariant statistic (`prereg_E2_identidad.md`): on the two halves of the probe, each head’s signature is recomputed with the same code as lemma 2, the matrix

$$M_{hh'} = |\langle \mathbf{r}_h^{(A)}, \mathbf{r}_{h'}^{(B)} \rangle|$$

is formed per layer, and the identification rate is read: the fraction of heads whose signature on half A has its own signature on half B as nearest neighbour. The null is a thousand permutations of the labels of half B per layer, with expectation $1/H$. The pre-registered verdict requires the mean rate over layers to exceed the null’s 99th percentile in all five seeds.

It is met with room to spare (table 11). At $P=1000$ with class-balanced halves the mean rate is 0.993 ± 0.007 against a null 99th percentile of 0.333, in all five seeds and with all twelve layers above 10/12; with optimal assignment between heads the rate is 1.000 in every layer and seed. The secondary curve in P runs from 0.981 at $P=250$ to 0.999 at $P \geq 2000$. With the class-block

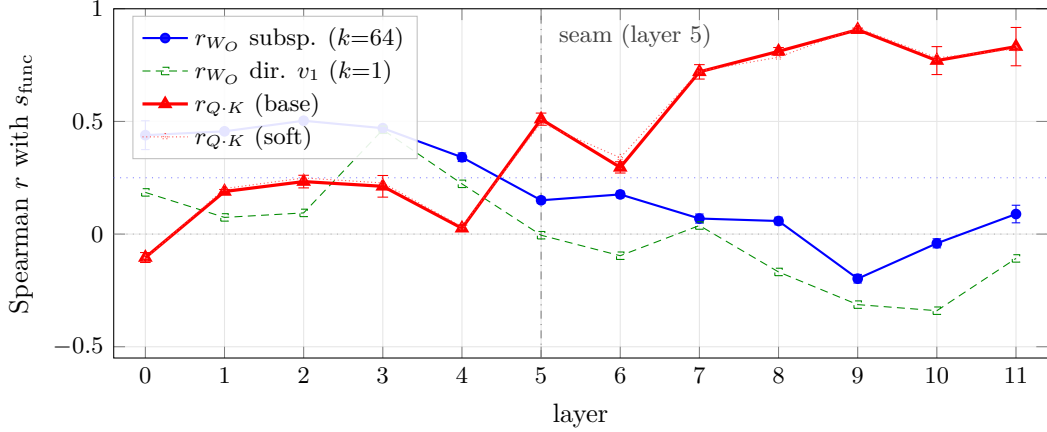


Figure 1: Premise, part II (visual of table 10): Spearman correlation between s_{func} and the geometric candidates per layer, mean across 5 seeds; bars are ± 1 standard deviation. The subspace of $\mathbf{W}_O$ ($k=64$) correlates in the early layers and sinks with depth. $Q \cdot K$ starts in the noise, switches on abruptly at the layer 5 *seam* ($0.026 \rightarrow 0.51$), and dominates up to ~ 0.9 . The soft variant’s $Q \cdot K$ (dotted) falls on top of base: the crossover does not depend on the weight regulariser. The dotted lines mark the random baselines (≈ 0.25 for the subspace, ≈ 0 for $Q \cdot K$).

halves of the published probe —the most adverse partition, which separates whole classes— the rate drops to 0.957 ± 0.006 , and the drop concentrates in layers 10 and 11 (0.77 and 0.80), where the signature depends most on class composition. The second part of the expectation is not met: the layers where the pruning decision disagrees across halves in the probe curve —layer 3, with two of five seeds coinciding, and layers 2 and 7— are not those with the lowest rate, which are 11, 2 and 8, and with the rate at 1.000 in nine of the twelve layers the ordering across layers is not informative. The test establishes the part the body needs: each head’s signature recognises itself across the probe, and that the pruning decision changes across halves in some layers is sensitivity of the *most redundant pair* to a small margin (lemma 2), not confusion between heads.

Table 11: Head identity across probe halves (5 anchor seeds; mean $\pm$ band). Identification rate with class-balanced halves at $P=1000$, rate under optimal assignment, 99th percentile of the permutation null (expectation $1/12$), and rate with the class-block halves of the published probe.

Layer	Rate (by class)	Optimal	Null p99	Rate (blocks)
0	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
1	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
2	0.983 ± 0.037	1.000	0.333	0.983 ± 0.037
3	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
4	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
5	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
6	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
7	1.000 ± 0.000	1.000	0.333	1.000 ± 0.000
8	0.983 ± 0.037	1.000	0.333	0.967 ± 0.046
9	1.000 ± 0.000	1.000	0.333	0.967 ± 0.046
10	1.000 ± 0.000	1.000	0.333	0.767 ± 0.091
11	0.950 ± 0.046	1.000	0.333	0.800 ± 0.095

B.3 The residual carrier tilts from inertness without decoupling

The gauge-invariant response signature (section 4) is defined on the contribution to the residual stream, not on $\mathbf{v}_1(\mathbf{W}_O)$. A prior diagnostic, without retraining, measures how far apart they are. Per layer, each head’s computed signature $\mathbf{v}_1(\mathbf{A}_h \mathbf{V}_h \mathbf{W}_O)$ is compared with its static signature

$\mathbf{v}_1(\mathbf{W}_O)$ —the *paired* angle—, against the permutation null —the computed signature of h versus the static one of $h' \neq h$ —. Both live in the 64-dimensional row space of $\mathbf{W}_O$, so chance here is the permutation within that subspace, not that of $\mathbb{R}^{768}$. Both signatures are computed by exact SVD.

The paired angle stays below the null at every layer and is never zero (table 12): the computed signature does genuinely tilt relative to the static one —value/attention content moves it, the residual carrier does not collapse onto the inert one, which was the go/no-go condition— but it remains more similar to its own head than to the rest, that is, head-specific. The gap is modulated by depth —from 24° at layer 1 to 39° at layer 2— with two irregularities reproducible across seeds: at layer 8 the null drops to 73.1° against the $\sim 86^\circ$ of the rest —the static directions $\mathbf{v}_1(\mathbf{W}_O)$ of different heads resemble each other there— and at layer 11 paired and null fall together (23.5° and 51.9°), with the whole pattern shifted downward. Both are properties of the static signature in the gauge that training fixed —not identifiable, like any $\mathbf{v}_1(\mathbf{W}_O)$ readout (section 7.1)—, so they are named as features of the diagnostic, not as redundancy or functional convergence. This is a diagnostic of the carrier, not the inversion of appendix B.1, and it is reported as such.

Table 12: Computed/static diagnostic of the residual carrier, without retraining (mean $\pm$ standard deviation across 5 seeds, frozen set of 1000 images). Paired angle (computed signature of h vs. static of h) against the permutation null (vs. static of $h' \neq h$), in degrees. Paired below null at every layer, and never at zero, indicates that the residual carrier tilts from inertness without decoupling.

Layer	Paired	Permutation null
0	43.0 ± 1.2	77.2 ± 0.2
1	60.4 ± 1.2	84.4 ± 0.1
2	46.6 ± 2.5	85.6 ± 0.1
3	50.8 ± 3.1	87.0 ± 0.1
4	50.8 ± 4.2	87.5 ± 0.1
5	59.5 ± 3.3	86.7 ± 0.2
6	52.3 ± 2.7	85.6 ± 0.2
7	64.4 ± 3.7	85.3 ± 0.2
8	44.2 ± 1.2	73.1 ± 0.7
9	49.6 ± 3.7	86.6 ± 0.2
10	52.3 ± 2.7	82.8 ± 0.5
11	23.5 ± 2.2	51.9 ± 1.6

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